

# The Anti-Competitive Effect of Input Tariff Liberalization\*

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## Abstract

This paper studies the welfare implications of input tariff liberalization in a model featuring firm heterogeneity in both exposure and responses to input tariff shocks. We show that this setting gives rise to a new anti-competitive effect that dampens the benchmark welfare gains from trade liberalization. We derive a sufficient statistic for this anti-competitive effect in a general environment: the correlation between firm market shares and their cost shares of imported inputs. The interplay between import intensiveness and market power implies that larger firms benefit more from input tariff cuts while passing through less of the cost reductions, leading to increased aggregate markup dispersion. We empirically test this framework using granular regulatory data from Colombia around its 2010 trade reform. The data covers the universe of firm-level inputs, both imported and domestically sourced, allowing us to disentangle firm-specific exposure and responses to input tariff shocks. Our quantitative analysis demonstrates that input tariff liberalization induces a substantial anti-competitive effect, comparable in magnitude to the pro-competitive effects in the studied episode.

*Keywords:* input trade liberalization; variable markups; anti-competitive effect

*JEL Classification:* F12; F13; O24

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# 1 Introduction

How large are the gains from trade through the competitive channel? Existing studies mostly focus on the pro-competitive effects of output tariff liberalization. The benchmark work of [Arkolakis et al. \(2012, hereinafter ACR\)](#) identifies the domestic expenditure share as a sufficient statistic for quantifying gains from trade under constant markups. Other seminal studies, such as [Edmond et al. \(2015\)](#) and [Arkolakis et al. \(2019, hereinafter ACDR\)](#), enrich the benchmark framework by allowing for variable markups to account for additional sources of gains from trade that result from reduced markup distortions due to fiercer competition from foreign exporters. Despite the coincidence of most output and input trade liberalization episodes, relatively little attention has been paid to the input tariff channel.

In this paper, we explore the competitive welfare effects of input tariff liberalization. In a standard oligopoly model setting ([Atkeson and Burstein, 2008](#)) where a firm optimally chooses its markup according to its domestic market share, we embed firm-specific input tariff shocks into the model to uncover a new channel of the anti-competitive effect of input tariff liberalization. We prove that the existence of the anti-competitive effect hinges on one commonly observed feature in firm level data. Firms with larger market shares tend to have higher import shares and smaller firms tend to exhibit lower import intensities.<sup>1</sup> Because large firms have higher import shares they benefit more from cheaper imported inputs. Large firms nevertheless raise their markups and do not fully pass through the input cost cuts. Small firms, in contrast to the large ones, have much poorer access to input importing and are thus far less exposed to input tariff cuts despite their willingness to keep markups constant and perfectly pass through the cost reductions. The interplay of exposure and responses at the micro-level aggregates up to a greater markup dispersion and muted welfare gains from trade. This anti-competitive effect of input tariff liberalization pulls against

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<sup>1</sup>In the context of this paper, we use “import share” and “import intensity” interchangeably. Both terms are defined as the value share of imported inputs over total input purchases. Firm market shares are within-sector.

the well-studied pro-competitive gains from trade of output tariff liberalization.

By decomposing the overall welfare effects into the pro- and anti-competitive effects resulting from output and input tariff liberalization, respectively, we derive an analytical formula of the change in welfare level before and after trade liberalization. We show that the anti-competitive effect exists if and only if firm market shares and firm import shares are positively correlated. In this case, incorporating the input tariff channel into the benchmark framework predicts lower welfare gains from trade than in standard settings without accounting for heterogeneous firm markup responses to input tariff shocks.

To empirically test our theoretical framework, we use the firm microdata in Colombia from its Annual Manufacturing Survey (Encuesta Anual Manufacturera (EAM)) around its 2010 trade reform. The EAM requires firms to report their value shares of each imported input and exported output among all 8-digit CPC product codes.<sup>2</sup> The richness of the dataset allows us to observe firm-level heterogeneity in both exposure and responses to input tariff shocks.

In year 2010, Colombia pushed through its Tariff Structural Reform (TSR) to promote imports of intermediate inputs.<sup>3</sup> Using the Colombian firm microdata before and after the trade reform, we observe two stylized facts. First, there is a positive correlation between firm market shares and import shares. Second, large firms, unlike small firms, increase their profit margins when benefiting from cheaper imported inputs upon input tariff liberalization. The synergy of the two — one on the heterogeneous exposure and the other on the responses — sheds light on the aforementioned mechanism of how the input tariff liberalization can lead to anti-competitive welfare effects on post-reform Colombia.

Our empirical results show: (a) that a given input tariff cut affects the sales of large firms more positively than small firms — before converting the tariff cut to a

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<sup>2</sup>The first five digits can be concorded to the UN standard 5-digit CPC codes and then to the HS 6-digit tariff data. The last three digits are specifically encoded within Colombia. We still leverage all the 8-digit entries for variations in firm-product input shares.

<sup>3</sup>Another agenda of TSR is to encourage imports of capital goods. [Meleshchuk and Timmer \(2020\)](#) study the same trade liberalization episode in Colombia but focus on the capital goods.

firm-specific input cost share,<sup>4</sup> and (b) that this interaction effect flips sign and remains significant after we convert input tariff shocks into firm-level cost shocks to take into account differences in import-intensiveness of a given input bundle. With this measure of the firm-specific exposure to input tariff reductions, our main results are consistent with a large class of models that predict that larger firms exhibit lower pass-through of own cost shocks.

We then feed the firm microdata into the model to analyze quantitatively the determinants of the anti- and pro-competitive effects and their relative magnitudes in the overall welfare gains from trade. Our analysis suggests that the magnitude of the anti-competitive effects can be on par with the pro-competitive effects in the Colombian trade liberalization episode.

Our paper is most related to three strands of literature: a) quantification of welfare gains from trade through the competitive channels; b) empirical evidence on input tariff liberalization and firm performance; c) firm performance in response to broader cost shocks.

ACDR (2019) provide a new ACR formula when allowing for markup endogeneity and underline two opposite competitive effects from output tariff liberalization: a) exporters' markup adjustment which can go in the opposite direction to the changes in domestic markups, and b) reallocation of market share to more productive firms due to selection. The overall competitive effects depend on the relative importance of the two. [Edmond et al. \(2015\)](#) study the pro-competitive gains from trade due to reduced markup distortion. [Hsu et al. \(2020\)](#) shows similar results using Chinese firm-level data. The pro-competitive effects of trade can lead to welfare losses if trade increases markup dispersion and hence amplify distortions ([Epifani and Gancia, 2011](#)). [Dhyne et al. \(2022\)](#) introduce network structure and simulate the aggregate effects of

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<sup>4</sup>This has been the common measure of input tariffs, i.e., the weighted average of import tariffs with weights being sectoral input-output coefficients, as firm-product-level input information is rarely observed. The measure implicitly assumes away firm heterogeneity in import-intensiveness, which is instead picked up by the coefficient. The results would misleadingly suggest that large firms are more responsive to cost reductions. In fact, large firms take market shares from small firms not because they pass through more input cost reductions and expand sales by more, but only because they are exclusively benefiting from the input tariff cuts.

uniform trade liberalization. They find that the welfare effects are similar under fixed and endogenous markups, and that the effects on markups of domestic firms are on average close to zero. Our paper adds to the existing literature by highlighting the competitive effects through the input trade channel. As opposed to the pro-competitive effects of output trade liberalization, we show that the input trade channel can lead to anti-competitive effects.

The second strand of the literature documents the empirical evidence on the impact of input trade liberalization. Copious studies find that imports of intermediates or declines in input tariffs are associated with sizable firm performance improvement ([Amiti and Konings, 2007](#); [Kasahara and Rodrigue, 2008](#); [Goldberg et al., 2010](#); [Halpern et al., 2015](#)). The most relevant ones to ours are [De Loecker et al. \(2016\)](#) and [Fan et al. \(2018\)](#). The authors point out the importance of incorporating the input tariff channel and document that firms raise markups when facing input tariff cuts. In this paper, we disentangle firm-level heterogeneity in both exposure and responses and go one step further to provide a framework for better understanding of the aggregate welfare implications.

A broader literature looks into the firm markup responses to cost shocks. [Berman et al. \(2012\)](#) study the exchange rate shocks and show that currency depreciation leads to enlarged market shares thus increased markups, more so for large firms. [Amiti et al. \(2019\)](#) find evidence of heterogeneity in markup variability across small and large firms in response to exchange rate shocks, in terms of both own cost pass-through and strategic complementarities. Our paper is closely linked to this strand of the literature yet with an emphasis on input cost shocks whose impact is differential across domestic firms through the input tariff channel.

The rest of the paper is organized as follows. Section 2 shows the theoretical framework of the anti-competitive welfare effects. Section 3 documents stylized facts and empirical evidence. Section 4 provides quantitative analysis. Section 5 concludes.

## 2 Theoretical Framework

In this section, we first describe the theoretical framework that allows for embedding of firm level heterogeneity in both market shares and import shares as observed in the data. Based on this framework we proceed to derive the overall welfare gains from trade and show the conditions for the existence of the anti-competitive effect.

### 2.1 Marginal Cost

We begin with the firm side and impose mild assumptions on the production structure. The marginal cost function of firm  $i$  is<sup>5</sup>

$$MC_i = \frac{1}{A_i} C_i \left( P_{fi}^m, P_d^m \right), \quad (1)$$

where  $A_i$  is a Hicks-neutral firm productivity shifter, and  $C_i \left( P_{fi}^m, P_d^m \right)$  is a function of firm  $i$ 's price indices for foreign inputs  $P_{fi}^m$  and domestic inputs  $P_d^m$ , which includes intermediates and labor.<sup>6</sup>

Based on this general marginal cost function, we use Shephard's Lemma to write the log change in firm marginal cost as<sup>7</sup>

$$dmc_i = z_i dp_{fi}^m + (1 - z_i) dp_d^m - da_i, \quad (2)$$

where small letters denote log transformation, and  $z_i$  is the import intensity of firm  $i$ , measured as the expenditure share on imported inputs in total expenditure including that on domestic and foreign intermediate inputs and the wage bill. Lower-case letters are logs.

We can further write out the imported input cost index as  $dp_{fi}^m = \sum_{s'} \theta_{i,s'} dp_{fs'}^m$

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<sup>5</sup>This and the following expressions applies to all periods and the time  $t$  subscript is suppressed. We will show the  $t$  subscript when moving to the firm panel data analysis.

<sup>6</sup>Note that the markup function implies constant returns to scale.

<sup>7</sup>While the following derivations are of first-order approximation, we write down a general equilibrium model to study the overall welfare effects. Our simulation results show that the first-order effects indeed captures the main effects and the second-order effects are negligible under imperfect competition. We leave these results in Appendix C so that the main body of the paper remains coherent in that we do not impose a specific firm production function.

with  $\theta_{i,s'}$  being firm  $i$ 's expenditure share of imported input  $s'$  in its total cost of imported inputs. Importing intermediate input  $s'$  incurs iceberg trade costs  $\tau_{s'}$ . During a trade liberalization episode,  $dp_{fs'}^m$  can be largely driven by changes in trade costs  $d\tau_{s'}$ . Now assume that the cost changes are mainly due to changes in tariffs. Hence the first component in the first-order expansion can be written explicitly as  $dp_{fi}^m = z_i \sum_{s'} \theta_{i,s'} d\log(1 + \tau_{s'})$ , which is a function of the tariff shocks  $d\tau_{s'}$ , and a firm's overall import intensity  $z_i$ , plus the composition of the firm's imported intermediate inputs  $\theta_{i,s'}$ . We map our model to the widely adopted Bartik-type construction of input tariff shocks in many existing empirical studies by defining input tariff shocks as the weighted average of output tariff shocks  $d\tau_i^{input} \equiv \sum_{s'} \theta_{i,s'} d\log(1 + \tau_{s'})$ , and simplify the first term as  $dp_{fi}^m = z_i d\tau_i^{input}$ .

As we will focus on the effects of trade shocks on firm marginal cost, we further simplify the change in firm marginal cost by assuming that there is one domestic intermediate producer who prices at CES markups and uses only labor. The intermediate good producer does not directly sell to consumers. Lastly, we assume constant productivity,  $da_i = 0$ . Therefore, the first-order expansion of the change in the marginal cost of domestic firm  $i$  can now be written down as

$$dmc_i = z_i d\tau_i^{input} + (1 - z_i) d\log w, \quad (3)$$

which depends on firm import share  $z_i$  and the firm-specific input tariff shock  $d\tau_i^{input}$ .

## 2.2 Prices

We take the price change decomposition from the [Amiti et al. \(2019\)](#) setup and write the first-order expansion of firm  $i$ 's price responses to its own cost shocks and to the price setting of its competitors within a given industry (i.e. strategic complementarities)<sup>8</sup>

$$dp_i = \frac{1}{1 + \Gamma_i} dmc_i + \frac{\Gamma_{-i}}{1 + \Gamma_i} \sum_{j \neq i}^N \frac{s_j}{1 - s_i} dp_j + \frac{\Gamma_{-i}}{1 + \Gamma_i} \frac{s_f}{1 - s_i} dp_f, \quad (4)$$

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<sup>8</sup>Note that this requires the demand system to be invertible, a rather mild assumption on demand that is compatible with common demand systems in the literature.

where  $s_i$  is firm  $i$ 's market share,  $s_f = 1 - \sum_i s_i$  is the share of foreign goods in the final goods market, and  $\Gamma_i$  and  $\Gamma_{-i}$  are the own and competitor markup elasticities, respectively.<sup>9</sup>

Rearrange the price terms to obtain

$$dp_i = \frac{1 - s_i}{1 - s_i + \Gamma_i} dmc_i + \frac{\Gamma_{-i}}{1 - s_i + \Gamma_i} d\log P_i, \quad (5)$$

where  $d\log P_i = s_i dp_i + \frac{\Gamma_{-i}}{\Gamma_i} \left( \sum_{j \neq i}^N s_j dp_j + s_f dp_f \right)$  is a weighted average of price change in the industry where the weights depend on firm  $i$ 's own-markup elasticity relative to its competitor markup elasticity.

Under the Cournot quantity competition with Cobb-Douglas between-industry demand, we have  $\Gamma_i = \Gamma_{-i} = (\sigma - 1)s_i$ , a monotonically increasing function of firm  $i$ 's market share  $s_i$ . The price change decomposition becomes

$$dp_i = \frac{1 - s_i}{1 + (\sigma - 2)s_i} dmc_i + \frac{(\sigma - 1)s_i}{1 + (\sigma - 2)s_i} dp. \quad (6)$$

The percentage change in the industry price index  $dp$  can be written out as the average of price change in the sector weighted by firm market shares

$$dp = \sum_{i=1}^N s_i dp_i + s_f dp_f. \quad (7)$$

Note that  $p$  can be viewed as the full price index in the sector as it incorporates both domestic and foreign competitors selling in the domestic final market.

Assuming that home is small and foreign firms do not play the cournot game, we simply have

$$dp_f = d\tau. \quad (8)$$

where  $\tau$  is the output tariff.<sup>10</sup>

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<sup>9</sup>This and the following derivations are all applied to firms in one industry, therefore the industry subscript  $s$  is dropped. We do allow for multiple sectors when moving toward empirics in the next section, and in the welfare analysis as shown in robustness results (see Appendix D). Specifically, there are 26 two-digit manufacturing sectors in the Colombian manufacturing firm data.

<sup>10</sup>Note that import intensity is irrelevant when it comes to exposure to output tariff shocks, so  $d\tau$  is not  $i$ -variant. In the empirical section, however, we do control for firm-specific output tariff shocks as we observe the output composition at the firm-product level. The model is so far silent on such firm

Combining equations (3), (6), (7), and (8), we are able to derive a closed-form solution for the change in real wage as shown in the next subsection.

## 2.3 Welfare Analysis

We proceed to analyze the welfare effects of trade liberalization by combining the above price equations. We derive an analytical result of the change in welfare level before and after the trade liberalization, and decompose the welfare effects into the pro- and anti-competitive effects due to output and input trade liberalization, respectively.<sup>11</sup>

**Proposition 1.** *When firms compete in the final goods market à la Cournot, as in the goods market environment described in Sections 2.1-2.2, the change in the price index is given by:*

$$\begin{aligned}
 dp = & \underbrace{s_f d\tau + \sum_{i=1}^N s_i z_i d\tau_i^{input}}_{\text{Gains without competitive effects}} + \underbrace{s_f d\tau \frac{\sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}}{1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}}}_{\text{Pro-competitive effect}} \\
 & + \underbrace{\frac{\text{Cov}\left(z_i, \frac{(\sigma-1)s_i}{1+(\sigma-2)s_i}; s_i \left(-d\tau_i^{input}\right)\right)}{1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}}}_{\text{Anti-competitive effect}} \quad (9)
 \end{aligned}$$

where  $\text{Cov}(a_i, b_i; c_i)$  is the weighted covariance between  $a_i$  and  $b_i$  with weights  $c_i$ .

*Proof.* See Appendix B.1. □

The welfare decomposition expression provides us with the foundation for analyzing the sufficient and necessary conditions for the existence of an anti-competitive effect, and for discussing its quantitative significance compared to the well-documented pro-competitive effect. Below we provide an intuitive interpretation of the welfare

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heterogeneity on the output side.

<sup>11</sup>We take labor as the numeraire in the single-sector model. We do not consider tariff revenue in the welfare formula, and the result should best be interpreted as the consumer welfare.

decomposition formula.

### 2.3.1 Gains Without Competitive Effects

If all firms are small relative to the market ( $s_i \rightarrow 0$ ), we converge to the full pass-through world as the second and the third term in the decomposition go towards zero. Both output and input tariff liberalization brings about welfare gains without considering variable markups.

Gains from output trade are governed by openness ( $s_f$ ) and the size of output tariff cuts ( $d\tau$ ), while gains from input trade depend on the size of input tariff cuts ( $d\tau^{input}$ ) and the correlation of firm market shares and import shares. Without incorporating the competitive effects, input tariff liberalization brings about greater welfare gains when large firms have higher import exposure as these firms take up higher weights in the aggregate (Blaum et al., 2018).

### 2.3.2 The Pro-Competitive Effects

During a trade liberalization episode, the second term is always negative given  $d\tau < 0$  and increases with: a) import competition governed by the foreign share  $s_f = 1 - \sum_i s_i$ , and b) rising concentration in the domestic sector, measured by  $\sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}$ .

To understand this term in an intuitive manner, we consider a special case as  $\sigma \rightarrow 2$ . Under this case, the pro-competitive effect term converges to:

$$\frac{HHI}{1 - HHI} s_f d\tau$$

where  $HHI = \sum_{i=1}^N s_i^2$ . The magnitude of the pro-competitive effect is increasing in the Herfindahl-Hirschman Index (HHI). The intuition is that pro-competitive effects are greater when we see higher concentration thus more misallocation to be corrected by the competition pressure from output tariff liberalization.

### 2.3.3 The Anti-Competitive Effect

The third term depends on the covariance of firm import intensity  $z_i$  and a positive function of firm market share  $\frac{(\sigma-1)s_i}{1+(\sigma-2)s_i}$ , weighted by the multiplication of a firm's market share and its exposure to input tariff shocks – if there is a positive correlation between the two, which corresponds to higher import intensity among larger firms, there will be a positive effect on the price index and hence a negative one on real wage. As this is more likely to be the empirically relevant case, one would expect this term to lead to an anti-competitive effect. As is the case with the pro-competitive effect, the magnitude of the potential anti-competitive effect is increasing in market concentration.

Consider again a special case with  $\sigma = 2$ , for which we can simplify the weighted covariance term into:

$$\frac{\text{Cov} \left( z_i, s_i; s_i \left( -d\tau_i^{\text{input}} \right) \right)}{1 - \text{HHI}}$$

It becomes clear in this special case that we have an anti-competitive effect **if and only if**  $\text{Cov} \left( z_i, s_i; s_i \left( -d\tau_i^{\text{input}} \right) \right) > 0$ , which means that the correlation of firm market share and import share is positive when weighted by  $s_i \left( -d\tau_i^{\text{input}} \right)$ . Moreover, the strength of the anti-competitive effect increases with HHI in the domestic sector. Beyond the special case, we have the following corollary on the existence of an anti-competitive effect:

**Corollary 1.** *Under Cournot competition, a necessary and sufficient condition for the existence of an anti-competitive effect of trade is given by<sup>12</sup>*

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<sup>12</sup>With  $\sigma > 1$ ,  $\frac{(\sigma-1)s_i}{1+(\sigma-2)s_i}$  is increasing in  $s_i$ . Given that, one may tend to think that  $\text{cov} \left( s_i^I, \frac{(\sigma-1)s_i}{1+(\sigma-2)s_i}; s_i \left( d\log w - d\tau_i^{\text{input}} \right) \right) > 0$  and  $\text{cov} \left( s_i^I, s_i; s_i \left( d\log w - d\tau_i^{\text{input}} \right) \right) > 0$  are equivalent. According to Lehmann (1966), however, we need a slightly stronger positive dependence between  $s_i$  and  $s_i^I$  to guarantee a positive correlation between any increasing function of  $s_i$  and  $s_i^I$ , which is  $(s_i^I, s_i)$  being positive quadrant dependent (PQD). PQD is commonly used as a benchmark as to the strength of positive dependence. While positive correlation is the weakest concept of positive dependence, PQD is slightly stronger. To be more specific,  $(X, Y)$  is positive quadrant dependence if  $\Pr(X \geq x, Y \geq y) \geq \Pr(X \geq x) \Pr(Y \geq y)$ . Lehmann (1966) showed that PQD is equivalent to  $\text{cov}(a(x), b(y)) \geq 0$  for any pair of increasing functions  $a$  and  $b$  defined on  $R$ . Both positive correlation and PQD belong to the family of positive dependence, which in our context means firm  $i$  having a large

$$Cov\left(z_i, \frac{(\sigma-1)s_i}{1+(\sigma-2)s_i}; s_i \left(-d\tau_i^{input}\right)\right) > 0. \quad (10)$$

The proof immediately follows from Proposition 1. Below is a case-by-case discussion on  $Cov\left(z_i, \frac{(\sigma-1)s_i}{1+(\sigma-2)s_i}; s_i \left(-d\tau_i^{input}\right)\right) > 0$  being both **necessary** and **sufficient** for the existence of the anti-competitive effect.

**Case 1.**  $Cov\left(z_i, \frac{(\sigma-1)s_i}{1+(\sigma-2)s_i}; s_i \left(-d\tau_i^{input}\right)\right) = 0$ .

We provide a formal proof in Appendix B.2 that markup dispersion remains unchanged in the case with symmetric exposure to input tariff shocks across firms. Here, we discuss intuitively the theoretical reason for this result. Now that the marginal cost shocks are identical across firms, we argue that large firms do not want to lose their market shares to small firms and end up having to keep their markup levels unchanged. It is helpful here to bring up one intermediate step in deriving the price equations:  $d\mu_{i,t} = -\Gamma_{i,t} (dp_{i,t} - dp_t)$  where  $\mu_{i,t}$  stands for firm  $i$ 's log markup in period  $t$ .<sup>13</sup> Suppose large firm  $i$  increases its markup level, we then have  $dp_{i,t} - dp_t > 0$  — large firm  $i$ 's relative price goes up as the price index goes down by more considering that small firms fully pass through the now symmetric shocks. Then back to the above markup equation — large firm  $i$  will face a downward pressure on its markup until  $dp_{i,t} - dp_t$  is no longer positive, that is,  $p_{i,t}$  does not decrease by less or large firm  $i$  does not pass through less. Hence, large firms would not be able to raise markup levels, and markup dispersion would remain unchanged.

In short, large firms (a) pass through less of the cost reduction and (b) react more to the decrease in the price index. In the case with no correlation between firm market and import shares, the two effects exactly balance out so that input tariff liberalization

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market share tends to coincide with it having a large import share. Therefore, a slightly stronger sufficient condition for the existence of an anti-competitive effect of trade is  $(s_i^I, s_i)$  being PQD. In practice, we find that  $cov\left(s_i^I, \frac{(\sigma-1)s_i}{1+(\sigma-2)s_i}; s_i \left(d\log w - d\tau_i^{input}\right)\right) > 0$  and  $cov\left(s_i^I, s_i; s_i \left(d\ln w - d\tau_i^{input}\right)\right) > 0$  are almost always equivalent out of 100,000 simulations under a reasonable range of  $\sigma$ 's and a large number of firms  $n > 100$ .

<sup>13</sup>Note that this is a first-order approximation result, and holds under a wide class of models including CES+MP, CES+oligopoly (Atkeson-Burstein), and non-CES (markups decreasing in demand elasticity).

would not lead to anti-competitive effects.<sup>14</sup>

**Case 2.**  $Cov\left(z_i, \frac{(\sigma-1)s_i}{1+(\sigma-2)s_i}; s_i \left(-d\tau_i^{input}\right)\right) > 0.$

In this case, large firms benefit more from cheaper imported inputs compared to small firms. Following a similar argument in Case 1, large firm  $i$  can now raise its markup level without worrying about losing its market share to small firms who do not enjoy input cost cuts of the same amount. There will not be the same downward pressure on firm  $i$ 's markup as long as its relative price is still going down. This is because large firms are now exclusively exposed to the input tariff cuts. This positive correlation between firm market and import shares leads to greater markup dispersion and dampens welfare gains from trade.

To summarize, large firms face less downward pressure on their markups relative to the upward incentive. The aforementioned effect (a) dominates effect (b). With a positive correlation between firm market shares and input shares, input tariff liberalization will lead to an anti-competitive effect.

**Case 3.**  $Cov\left(z_i, \frac{(\sigma-1)s_i}{1+(\sigma-2)s_i}; s_i \left(-d\tau_i^{input}\right)\right) < 0.$

In the opposite case with small firms having higher import shares (albeit inconsistent with most empirical findings), we flip the argument in case 2. The aforementioned effect (b) dominates effect (a) and large firms face even more downward pressure on their markups. This leads to a lower markup dispersion and an additional pro-competitive effect of input tariff liberalization.

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<sup>14</sup>The mechanism here differentiates from that in [Berman et al. \(2012\)](#) in which depreciation leads to enlarged market shares thus increased markups of exporters, the more so the larger the firms. Large firms adjust markups in response to exchange rate shocks by more than small firms because large firms perceive a lower demand elasticity and charge higher markups. In our paper, however, large firms face competition pressures from small firms when they are equally exposed to input tariff cuts. Therefore, it is an essential element in our paper that there is firm heterogeneity not only in terms of the markup elasticities with respect to the shocks but also in terms of the exposure to the trade shocks themselves.

### 3 Empirical Evidence

In this section, we document important stylized facts during a recent trade liberalization episode in Colombia and estimate the heterogeneous responses of large and small firms to input tariff cuts. We observe two layers of firm heterogeneity: one in terms of the exposure to the input tariff shocks, and the other in terms of their responses.

#### 3.1 Data

We use a firm microdata in Colombia from 2008-2015 which allows us to observe firm input and output information around its 2010 trade liberalization.<sup>15</sup> The EAM dataset is particularly well-suited to our paper in that it contains information on physical quantities and unit values of all products a firm produces and all inputs it uses. For each input (output) entry, we observe the value share of imported inputs (exported outputs) across all 8-digit CPC product codes that concord to 6-digit HS codes. EAM also contains standard firm-specific variables including sales, value added, investment, employment, wage bills, and more.

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<sup>15</sup>The EAM data dates back to the 1980s and updates on an annual basis. We did not want to cross the Global Financial Crisis in 2008 or extends to after 2015 when additional reforms kicked in that increased tariffs for locally produced raw materials and capital goods while lowering the rates for non-produced goods; this policy has been maintained since PIPE 2.0.

Table 1: Summary Statistics

| Variable                | Sample | Mean   | SD      | p5    | p10    | p50    | p90     | p95     | N      |
|-------------------------|--------|--------|---------|-------|--------|--------|---------|---------|--------|
| Sales                   | All    | 12,038 | 33,996  | 213   | 313    | 1,822  | 30,475  | 68,239  | 73,815 |
|                         | Small  | 3,220  | 14,507  | 204   | 296    | 1,531  | 13,606  | 73,325  | 67,571 |
|                         | Large  | 77,326 | 82,037  | 9,692 | 13,446 | 60,785 | 157,943 | 207,138 | 6,244  |
| Materials               | All    | 10,747 | 116,358 | 20    | 72     | 697    | 14,414  | 39,640  | 73,798 |
|                         | Small  | 4,826  | 37,443  | 18    | 66     | 582    | 6,335   | 14,051  | 67,556 |
|                         | Large  | 74,841 | 374,740 | 1,686 | 4,032  | 26,401 | 145,094 | 236,883 | 6,242  |
| Materials<br>(Imported) | All    | 2,156  | 28,618  | 0     | 0      | 0      | 725     | 4,689   | 73,798 |
|                         | Small  | 489    | 4,552   | 0     | 0      | 0      | 131     | 1,178   | 67,556 |
|                         | Large  | 20,200 | 95,414  | 0     | 0      | 815    | 42,189  | 80,414  | 6,242  |
| Import Intensity        | All    | 0.068  | 0.197   | 0     | 0      | 0      | 0.253   | 0.591   | 71,383 |
|                         | Small  | 0.052  | 0.174   | 0     | 0      | 0      | 0.118   | 0.466   | 65,272 |
|                         | Large  | 0.243  | 0.309   | 0     | 0      | 0.066  | 0.772   | 0.892   | 6,111  |

Note: Summary statistics of main firm characteristics using the final sample of EAM. The data span the period 2008 to 2015. Nominal variables are expressed in million of Colombian pesos (1 million COP  $\approx$  500 USD in year 2008). A firm is large if it belongs to the top quartile sectoral market share the first time it enters the sample and small otherwise. Materials stand for values of intermediate inputs. Materials (imported) only include the imported inputs. Firm import intensity is defined as its value share of imported inputs over total inputs.

Table 1 shows the summary statistics of the Colombia firm dataset with the main firm-level variables including sales, purchases of materials, and purchases of imported materials. Large firms, compared to small firms, have higher sales, purchase more intermediate products, and disproportionately use imported inputs more intensively. The last three rows display import intensities calculated as the ratio of imported materials across large and small firms. In both subsamples, most firms do not import. This is especially true for the small firm sample. On average, small firms source 5.2% of intermediate products from abroad. The number is much higher at 24.3% for large firms.

## 3.2 Policy Background

Since the 1980s, Colombia has established several commercial agreements with Costa Rica, Chile, Cuba, Mexico, Nicaragua, Panama, and the Bolivarian Republic of Venezuela in the framework of the Latin American Integration Association (LAIA) (ALADI, 2022). Additionally, for several decades, Colombia's tariff policy was determined by its membership in the Andean Community of Nations (CAN). Compliance with the CAN's Common External Tariff (CET) was no longer mandatory in 2008, allowing Colombia greater flexibility in trade policy from 2008 onward.

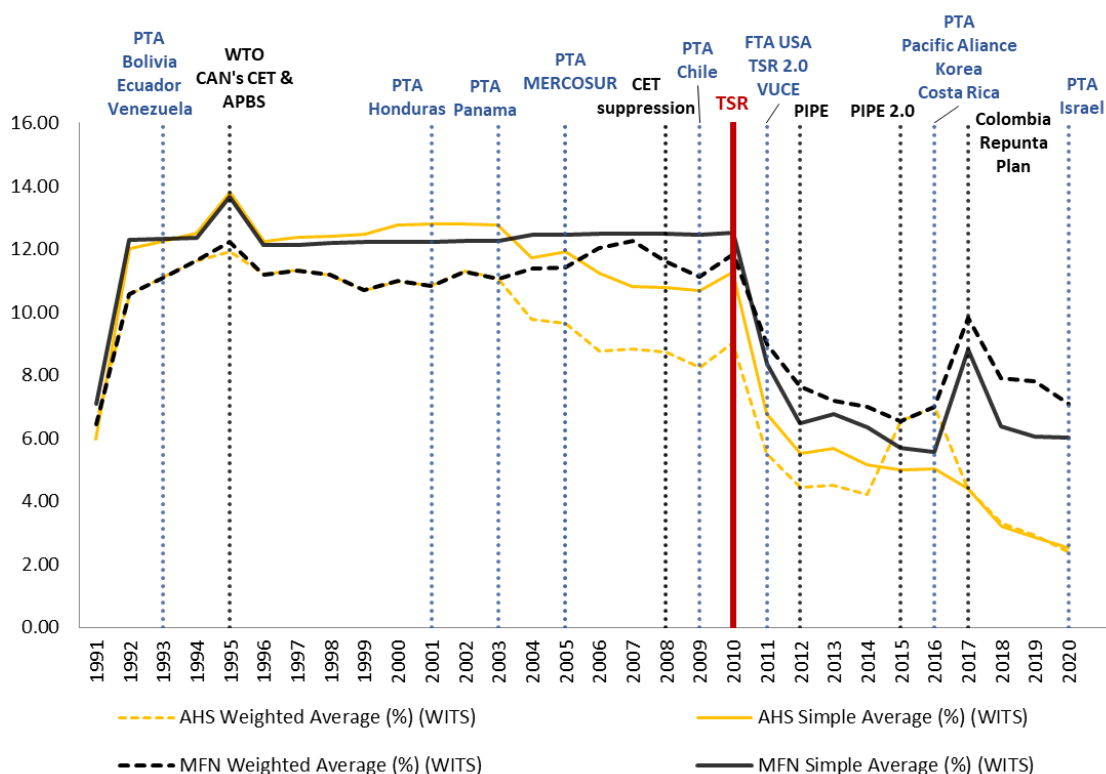


Figure 1: Tariff rates, Timeline, All Products, in Percentage Points. Note: CAN - The Community of Andean Nations; CET - The Common External Tariff; APBS - The Andean Community's Price Band System; PTA - Preferential Trade Agreement; MERCOSUR - Mercado Común del Sur (The Southern Common Market); TSR - Tariff Structural Reform; FTA - Free Trade Agreement; VUCE - Ventanilla Única de Comercio Exterior (Foreign Trade Single Window); PIPE - Plan de Impulso a la Productividad y el Empleo (Productivity and Employment Promotion Plan). Events colored in black impact multi-lateral tariff rates and events in blue affect only bi-lateral tariff rates.

The year 2010 is considered a breakpoint during which Colombia implemented a Tariff Structural Reform (TSR). The TSR was designed with several objectives, including reducing dollar supply pressures on the exchange rate, supporting the competitiveness of Colombian industry in international markets by reducing the cost of tradable inputs, and simplifying the tariff structure. The TSR implemented a tariff reduction targeted at imported raw materials and capital goods used for domestic production. The strategy consisted of gradually boosting the competitive capacity of local entrepreneurs and thus creating more jobs by reducing their costs and transforming the national market into an international one. On average, the MFN tariff rate went down from around 12% in 2010 to 8% in 2011, and further down to 6% in 2015. Figure 1 plots the timeline of Colombia tariff rates against its major tariff-related events.

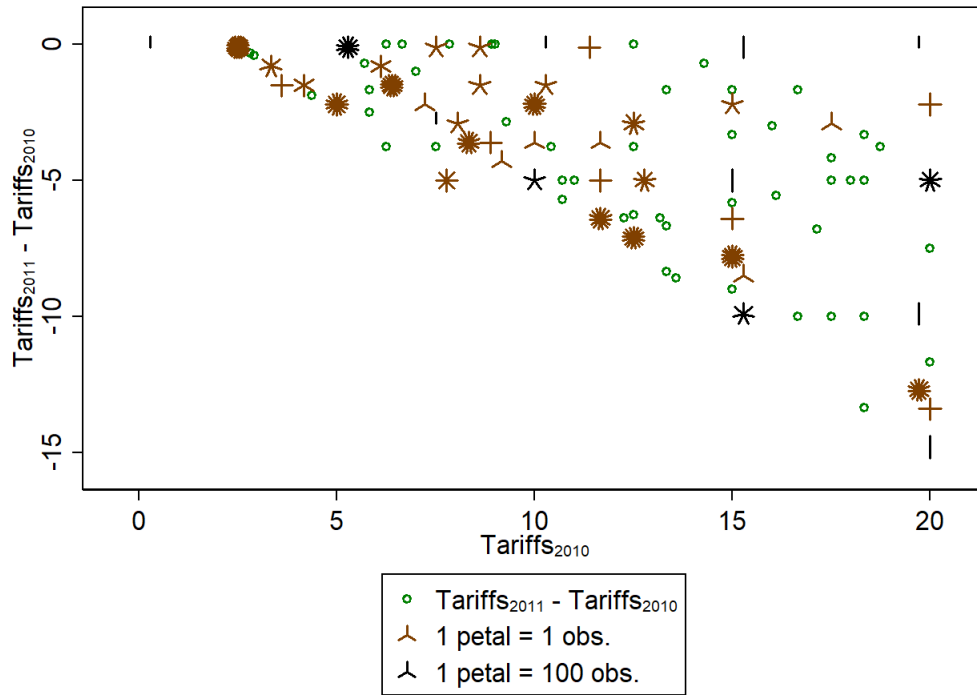


Figure 2: Colombia Trade Structural Reform, 2010-2011. Note: tariff cuts calculated using the WTO TAO tariffs across ~5000 HS 6-digit products.

The 2010 trade reform in Colombia contains enough variations both over time and across sectors. Figure 2 plots the tariff cuts against the initial tariffs. [Meleshchuk](#)

and Timmer (2020) argue that the trade reform can be treated as a “quasi-natural experiment” since the sectoral tariff cuts are highly correlated to the initial tariff rates. This suggests that some sectors underwent greater tariff cuts because they were previously more protected. Thus, the trade reform is aimed at reducing tariff dispersion rather than targeted at specific goods. We will deal with the potential endogeneity of tariff cuts when we proceed to the IV estimation.<sup>16</sup>

### 3.3 Stylized Facts

Using data on Colombian firms during the recent episode of trade liberalization, we document two stylized facts we observe in the Colombian firm microdata, which serves as the motivation of the paper.

**Fact 1.** *Firm import shares and market shares are positively correlated.*

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<sup>16</sup>Pilar Esguerra argues that the 2010 reduction in tariffs was related to the demands of some exporting guilds to fight against the negative effect that the appreciation of the peso was having on them. It is also worth mentioning that the 2010 trade liberalization episode overlaps with the Juan Manuel Santos government (2010-2018), during which there were several changes in the Colombian trade policy. The Santos administration was characterized by using trade as a strategy of global insertion and foreign policy to create links with several countries worldwide (Vargas-Alzate et al., 2012). Interestingly, the earlier major trade reform in the 1990s was when Santos got appointed the first minister of foreign trade in Colombia, who claimed to be “responsible for inserting the country to the world economy” (<https://www.rockefellerfoundation.org/profile/juan-manuel-santos/>).

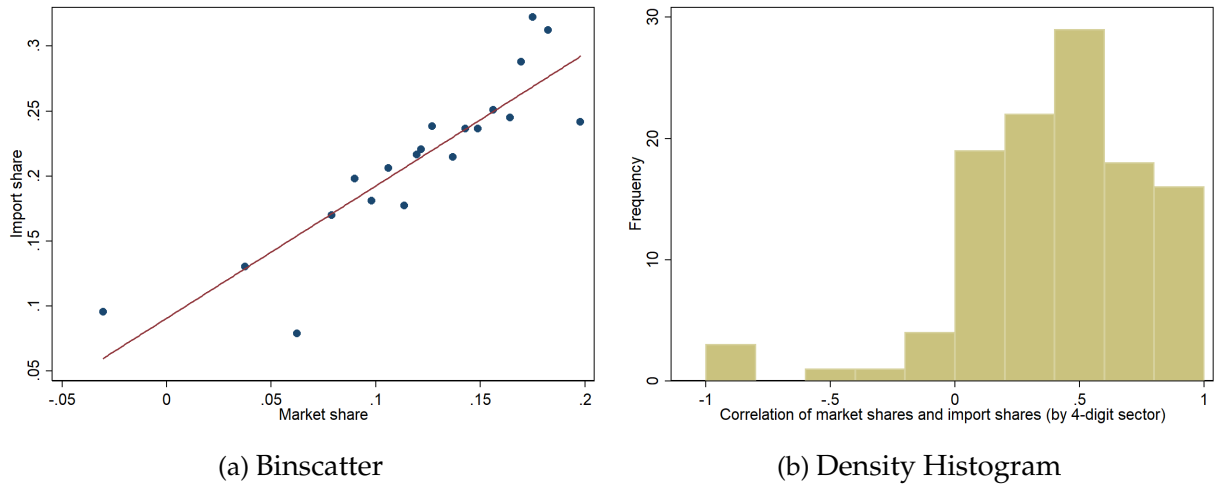
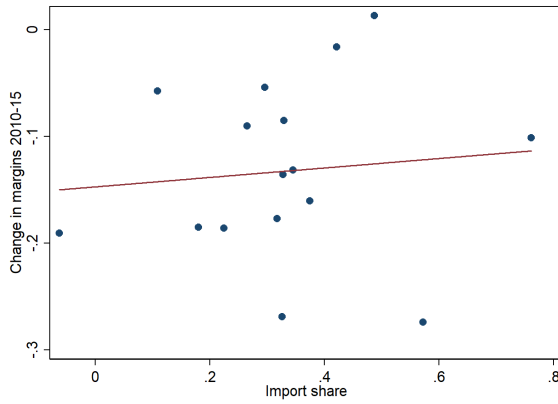


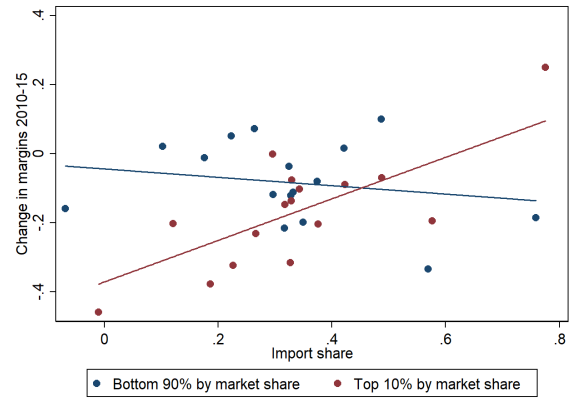
Figure 3: Firm Import Shares and Market Shares. Note: correlation calculated using the 2010 firm data right before the trade reform took place. The scatter plot includes four-digit ISIC sector fixed effects and is weighted by firm market share.

Figure 3a plots firm market shares against firm import shares and show the positive correlation of the two. Similar to what have been documented in many other countries, importers in Colombia are on average large firms, and large firms tend to be importers. In figure 3b, we further show the distribution of the correlation of firm market share and import share calculated for each 4-digit ISIC sector. The correlation distribution reinforces what we observe in the summary statistics, and indicates that the strength of the correlation also varies by sector.

**Fact 2.** *Import-intensive firms exhibits increasing margins since the trade reform; the effect is driven mainly by large firms.*



(a) All Firms, Binscatter



(b) Top 10% Firms versus bottom 90% by Market Share, Binscatter

Figure 4: Changes in Firm Profit Margins from 2010 to 2015 Plotted against Firm Import Share, Binscatter. Note: firm profit margins calculated as the firm revenue to cost ratio using the 2010 firm data right before the trade reform took place. Both plots include the subsample of importers and four-digit ISIC sector fixed effects and are weighted by firm market share.

Figure 4a and 4b show the changes in firm profit margins against firm import share.<sup>17</sup> While we observe an overall increase of firm margins in firm import shares, the pattern is solely driven by large firms. This pattern suggests that it is the synergy of high import share and high market share that leads to the changes in firm profit margins.

### 3.4 The Reduced-Form Regression

We run the following regression to capture the heterogeneous firm responses to both input and output tariff shocks, based on equations (6) and (8):<sup>18</sup>

<sup>17</sup>At this stage, we do not yet estimate firm-level markups to avoid assumptions on production function and parameter values. The firm profit margin measure serves well as our motivating facts.

<sup>18</sup>The baseline regression can be easily extended to account for multi-product firms. We prefer the firm-level to the firm-product-level regressions because firm inputs are reported as used by the firm as a whole but not by each product the firm produces. Firms can practically use the higher profits earned in one product line to subsidize and expand another one. The firm-product-level regressions will not add to variations in the explanatory variables and can potentially contaminate the response estimation.

$$\Delta y_{i,t} = \beta_1 \Delta \tau_{i,t}^{input} \times \text{Big}_i + \beta_2 \Delta \tau_{i,t}^{input} + \beta_3 \Delta \tau_{i,t}^{output} \times \text{Big}_i + \beta_4 \Delta \tau_{i,t}^{output} + \beta_{i(s),t} + \varepsilon_{i,t} \quad (11)$$

where the outcome variable  $\Delta y_{i,t}$  is firm  $i$ 's sales growth at year  $t$ , and  $\Delta \tau_{i,t}^{input}$  and  $\Delta \tau_{i,t}^{output}$  stand for the input and output tariff shocks faced by firm  $i$  at year  $t$ , and  $\beta_{i(s),t}$  are the sector-year fixed effects, and  $\text{Big}_i$  is a time-invariant dummy variable indicating if firm  $i$  is large the first time it enters the sample.<sup>19</sup>

Firm  $i$ 's exposure to input tariff shocks is constructed as a weighted average:  $\Delta \tau_{i,t}^{input} = \sum_{g \in G^I(i)} \theta_{i,g,t}^I \Delta \log(1 + \tau_{g,t})$ , where  $G^I(i)$  is the set of imported inputs of firm  $i$ ,  $\tau_{g,t}$  is the import tariff on product  $g$ , weighted by  $\theta_{i,g,t}^I$ , the value share of input  $g$  in total input cost of firm  $i$ . Similarly, firm  $i$ 's exposure to output tariff shocks is constructed as a weighted average:  $\Delta \tau_{i,t}^{output} = \sum_{g \in G^O(i)} \theta_{i,g,t}^O \Delta \log(1 + \tau_{g,t})$ , where  $G^O(i)$  is the set of outputs of firm  $i$ , i.e., products firm  $i$  produces and sells, and  $\tau_{g,t}$  is again the import tariff on product  $g$ , weighted by  $\theta_{i,g,t}^O$ , the value share of output  $g$  in total sales of firm  $i$ . The firm-specific input-output shares are observable at 8-digit CPC in EAM.

Table 2 shows the results of the baseline regression. Column 1-4 show the results using the "naive" value shares  $\theta_{i,g,0}^I$ , the pre-sample value share of input  $g$  in *total* input cost of firm  $i$ . Columns 5-8 switch to the "actual" value shares  $\theta_{i,g,t}^I$ , the contemporaneous value share of *imported* input  $g$  in total input cost of firm  $i$ . The responses of large firms to input tariff shocks relative to small firms are basically flipped when using the "naive" input tariff measure (using the overall input-output shares) versus the "actual" input tariff measure (taking into account firm-level import intensities). This shows that large firms expand relative to small firms not because they lower price and sell by more but only because they benefit more from input tariff shocks. Using the "naive" measure that implicitly assumes all firms import, the coefficient on input tariffs is picking up how large firms take advantage of the cheaper imported inputs to

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<sup>19</sup>In the baseline estimation, we define big firms as those in the top decile of the sectoral market share distribution the first time firms enter the sample. We explore alternative definitions of large firms, and the results remain qualitatively unchanged.

Table 2: Coefficient Heterogeneity, by Large and Small Firms, OLS Estimation

| Input Tariff Measure Using                                  |         | “naive” input tariff measure |                      |                     |                    | “actual” input tariff measure |                    |                    |                    |
|-------------------------------------------------------------|---------|------------------------------|----------------------|---------------------|--------------------|-------------------------------|--------------------|--------------------|--------------------|
| Dependent Variable:<br>$\Delta \ln(\text{Sales}_{i,t})$     | Sample: | (1)                          | (2)                  | (3)                 | (4)                | (5)                           | (6)                | (7)                | (8)                |
|                                                             |         | OLS<br>All                   | OLS<br>All           | OLS<br>Importers    | OLS<br>Importers   | OLS<br>All                    | OLS<br>All         | OLS<br>Importers   | OLS<br>Importers   |
| $\Delta \tau_{i,t}^{input}$                                 |         | 0.125<br>(0.106)             | 0.264**<br>(0.106)   | 0.688***<br>(0.242) | 0.581**<br>(0.269) | 0.380*<br>(0.229)             | 0.264<br>(0.244)   | -0.238<br>(0.259)  | -0.371<br>(0.281)  |
| $\Delta \tau_{i,t}^{input} \times \text{Large Firm Dummy}$  |         | -0.318**<br>(0.126)          | -0.591***<br>(0.155) | -0.510**<br>(0.252) | -0.485<br>(0.295)  | -0.0785<br>(0.299)            | -0.136<br>(0.316)  | 0.766**<br>(0.341) | 0.796**<br>(0.365) |
| $\Delta \tau_{i,t}^{output}$                                |         |                              | -0.00759<br>(0.0486) |                     | 0.0565<br>(0.0954) |                               | 0.0203<br>(0.0414) |                    | 0.0351<br>(0.124)  |
| $\Delta \tau_{i,t}^{output} \times \text{Large Firm Dummy}$ |         |                              | 0.134*<br>(0.00371)  |                     | 0.0549<br>(0.125)  |                               | 0.0919<br>(0.0702) |                    | 0.126<br>(0.132)   |
| Obs.                                                        |         | 43,708                       | 37,363               | 8,956               | 8,094              | 57,345                        | 47,608             | 9,706              | 8,781              |
| Sector $\times$ Year FEs                                    |         | yes                          | yes                  | yes                 | yes                | yes                           | yes                | yes                | yes                |

Note: OLS estimation using the final sample of EAM firm panel from 2008 to 2015. Column 1-4 using the “naive” input tariff measure, i.e. weighted averages in input tariff shocks where weights are firm-specific value shares of inputs over total purchase of inputs; column 5-8 using the “actual” input tariff measure where weights are firm-specific value shares of imported inputs over total purchase of inputs. All regressions include sector  $\times$  year fixed effects and firm fixed effects. Standard errors are clustered at the firm level. \*\*\*1%, \*\*5%, and \*10% significance levels.

steal market shares from small firms.<sup>20</sup> The EAM data enables us to use the “actual” measure to disentangle the actual changes in firm marginal costs from their responses as a result of the input tariff cuts. Controlling for firm-level heterogeneous exposure to such shocks reveals that large firms do not respond by as much as small firms do when they face the same own cost shocks.

We proceed with an instrument variable design to address the strategic targeting of tariff cuts following Faber (2014). We extract from the full list of 8-digit input product codes the most commonly used ones in terms of frequency and re-construct the input tariff measure for each firm. The instrument captures a significant portion of the “actual” tariff changes while being less likely to be influenced by unobserved firm-year-specific characteristics, e.g. firm lobbying. Table 3 shows the IV estimation results. Both the full-sample and sub-sample results are broadly consistent with a large class of model: small firms are more responsive to their own cost shocks (input tariffs) and large firms show stronger strategic complementarities (output tariffs).<sup>21</sup> One needs to be careful when interpreting the magnitude of the coefficients on input tariff shocks. The coefficients are economically huge due to the small (but not weak) coefficient in the first stage, which is almost mechanic due to the construction of the IV.<sup>22</sup> For an average small firm, a 1% decrease in input cost will lead to an increase in sales by 9% times its import intensity.<sup>23</sup> When it comes to the large firms, their responses to input tariff changes are neither statistically nor economically significant compared to the small ones.<sup>24</sup>

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<sup>20</sup>Note that the “naive” measure is commonly used in the literature to study the effect of input tariffs shocks, mainly due to the lack of firm-level input data. The fact that alternative measures of inputs tariffs can lead to completely opposite interpretations may explain the noisiness in the coefficient on input tariffs documented in the literature (for example, see De Loecker et al. (2016)).

<sup>21</sup>We use subgroups instead of the interaction term to allow for heterogeneous sector-time fixed effects as these fixed effects absorb the changes in the industry price indices to which firms of different sizes respond differently. More importantly, we do not find justification for using an instrumented interaction term with a dummy variable regarding firm characteristics (whether the definition of large firms are based on firm sales or employment).

<sup>22</sup>Magnitude-wise, the instrumented variable  $\approx$  the instrument  $\times$  import intensity.

<sup>23</sup>Note that the tariff cuts should be interpreted as percentage changes in input cost but not as changes in percentage points of tariff rates due to the construction of the tariff measure in log transformation. This facilitates the connection of the model to the reduced-form.

<sup>24</sup>Despite our focus on how firms respond to input tariff shocks, we add controls for output tariff shocks and find that large firms exhibit statistically and economically significant response to output

Table 3: Coefficient Heterogeneity, by Large and Small Firms, IV Estimation

| Large Firm Def.:                    | # Employees $\geq 100$ |                     |                     |                     | Top 10% Market Share |                     |
|-------------------------------------|------------------------|---------------------|---------------------|---------------------|----------------------|---------------------|
| Sample:                             | All                    | All                 | Small               | Large               | Small                | Large               |
| Dependent var.:                     | (1)                    | (2)                 | (3)                 | (4)                 | (5)                  | (6)                 |
| $\Delta \ln(\text{Sales}_{i,t})$    | IV                     | IV                  | IV                  | IV                  | IV                   | IV                  |
| $\Delta \tau_{i,t}^{\text{input}}$  | -5.553**<br>(2.756)    | -7.148*<br>(3.697)  | -8.983**<br>(4.570) | -3.411<br>(5.654)   | -9.051**<br>(4.437)  | -3.957<br>(5.183)   |
| $\Delta \tau_{i,t}^{\text{output}}$ | 0.301***<br>(0.073)    | 0.325***<br>(0.087) | 0.111<br>(0.114)    | 0.389***<br>(0.117) | 0.249**<br>(0.097)   | 0.578***<br>(0.199) |
| Obs.                                | 8,413                  | 8,409               | 4,066               | 4,327               | 5,639                | 2,762               |
| 1 <sup>st</sup> Stage F-Stat        | 44.2                   | 25.5                | 13.9                | 15.5                | 21.0                 | 6.4                 |
| Sector & Year FEs                   | Yes                    | No                  | No                  | No                  | No                   | No                  |
| Sector $\times$ Year FEs            | No                     | Yes                 | Yes                 | Yes                 | Yes                  | Yes                 |

Note: IV estimation using the final sample of EAM firm panel from 2008 to 2015. Results are 2nd stage IV estimates after instrumenting for input tariff changes. The specification is exactly identified. The instrument is based on the composition of imported inputs, i.e. weighted averages of input tariff changes where weights are firm-specific value shares of imported inputs over total purchase of imported inputs, fixed at the initial period. In constructing the weights, we only keep the most commonly used input product codes that belong to the top quantile across all Colombian firms in terms of input frequencies. Column 1 includes sector and year fixed effects; column 2-6 include sector  $\times$  year fixed effects. The definition of large firm is based on employment in columns 3-4 and on firm's sectoral market share the first time it enters the sample in columns 5-6. Standard errors are clustered at the firm level. \*\*\*1%, \*\*5%, and \*10% significance levels.

For robustness, we use alternative definitions of large firms. Column 3-4 of table 3 corresponds to the results using firm employment and column 5-6 are based on firm sectoral market share. We tried alternative cutoffs for the set of most commonly used inputs (>50%, 75%, 90%, etc.) and alternative definitions for large firms (top quantile, quintile, decile, etc.). The results remain qualitatively unchanged.

tariff shocks, consistent with models that predict stronger strategic complementarities of large firms. The opposite is true for small firms.

## 4 Quantitative Analysis

In this section, we quantify the aggregate implication on the overall welfare effects and the relative magnitude of pro- and anti-competitive effects by combining the welfare decomposition equation (9) with the microdata on the distribution of firm market shares and import shares.<sup>25</sup>

### 4.1 One-Sector Analysis

We loop over a set of reasonable values of the elasticity of substitution  $\sigma$  and of foreign goods shares in the domestic final goods market  $s_f$  and compare the resulting welfare consequence to test the sensitivity of our welfare estimation to alternative parameter values.

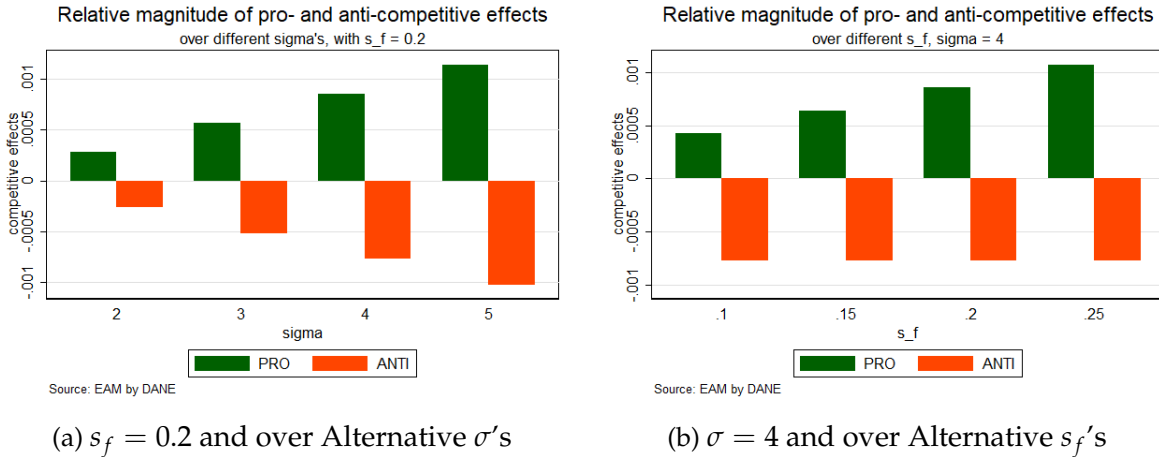


Figure 5: Relative Magnitudes of Pro- and Anti-Competitive Effects. Note: calculated using the 2010 firm data right before the trade reform took place.

Figure 5a shows the decomposed welfare effects over a range of  $\sigma$  from 2 to 5 and under a fixed share of foreign goods in the final goods market  $s_f = 0.2$ . In the

<sup>25</sup>Note that the relative magnitudes are governed by the pre-reform distribution of firm market shares and import shares plus the parameters ( $\sigma$  and  $s_f$ ), but are independent of the relative sizes of the output and input tariff cuts. Therefore, the comparison result is not a pure coincidence due to the nature of the trade liberalization episode in Colombia. One can think of these numbers as the pro- and anti-competitive coefficients and multiply them by a set of possible tariff cuts to obtain counterfactual welfare effects.

setting of the Colombian firms, the existence of the anti-competitive effect does not fully cancel out the pro-competitive effect, it wipes out around 80% of the latter. Since the sign of the anti-competitive effect is solely determined by the weighted covariance of firm market share and import intensity, the value of  $\sigma$  does not affect the existence of the anti-competitive effects. It is no surprise that the estimated welfare effects are qualitatively consistent over different  $\sigma$ 's.

Figure 5b shows the results over different levels of foreign competition under a fixed  $\sigma = 4$ . As foreign competition becomes fiercer, the pro-competitive effect starts to outweigh the anti-competitive effects, and vice versa. When the country is not facing enough foreign competition, which can be the case with home bias, the anti-competitive effect can dominate pro-competitive effect. In such cases, omitting the anti-competitive effect can lead to substantial overestimation of the aggregate welfare gains from trade.

## 4.2 Multi-Sector Analysis

We now extend the welfare quantification exercise in 4.1 to allow for multiple sectors. In Figure 6a, we plot out the distribution of the pro- and anti-competitive effects and the overall competitive effects for all 4-digit ISIC sectors.<sup>26</sup> While the median of the anti-competitive effect is smaller than the pro-competitive effect, its upper bound can be much higher due to high correlation of firm market shares and import shares in certain sectors. For 1/4 of the sectors, the anti-competitive effect dominates the pro-competitive effect.

Figure 6b shows a scatter plot of the pro-competitive effects against the anti-competitive effects for all 4-digit ISIC sectors.<sup>27</sup> The anti-competitive effect tends to be

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<sup>26</sup>For ease of display, we inverse the sign the the anti-competitive effect before comparing the magnitudes. It is worth mentioning that the anti-competitive effect term can yield an additional pro-competitive effect ("anticompetitive gains") in a small portion of sectors where firm market shares and import shares are negatively correlated. This is not observed in the single sector analysis when we aggregate over all firms across all sectors.

<sup>27</sup>Here, we drop sectors with "anticompetitive gains" in order to compare the absolute values of pro- and anti-competitive effects by sector.

stronger in industries where the pro-competitive effect is stronger. As predicted by our model, market concentration has a common role to play in amplifying both pro- and anti-competitive effects.

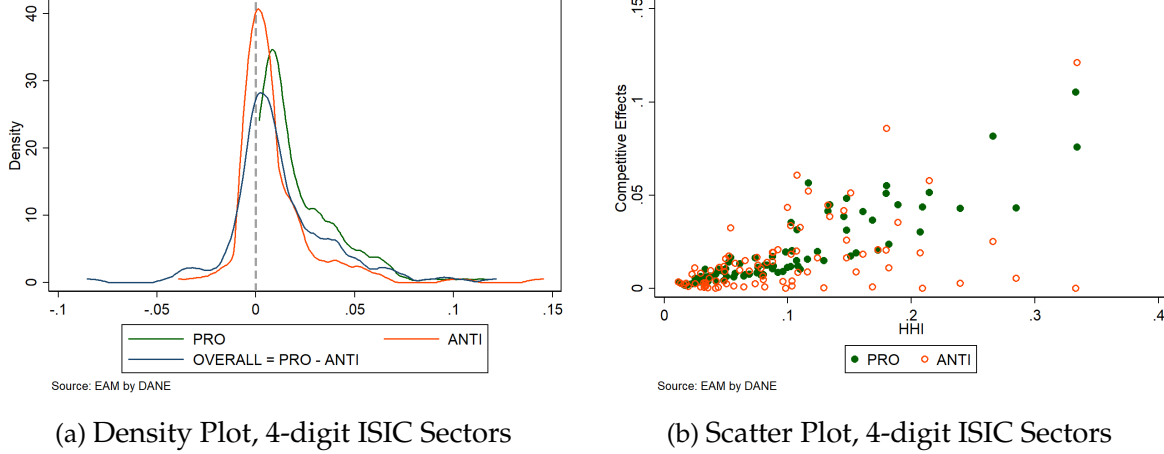


Figure 6: Relative Magnitudes of Pro- and Anti-Competitive Effects. Note: calculated using the 2010 firm data right before the trade reform took place. Parameter  $s_f$  calibrated using OECD Input-Output Tables (IOTs), and the multi-sector  $\sigma$ 's using Broda and Weinstein (2006).

## 5 Concluding Remarks

Our results contribute to the literature on firms' heterogeneous responses to own cost shocks, and specifically to input tariff shocks. To our knowledge, our paper is the first one to disentangle firm-level exposure and responses to input tariff shocks using a rich firm microdata that spans over a major trade reform in the country. Without observing firm-specific exposure, the regression coefficient on input tariff cuts misleadingly suggests that large firms would expand relative to small firms as the coefficient absorbs the differential import intensities across large and small firms. Large firms steal market shares from small firms during input trade liberalization not because they cut prices and sell more but because they are the ones who benefit more from input cost cuts. To account for the firm-varying import intensities, we instead measure input tariff shocks precisely at the firm level and use an instrumental variable design to address common endogeneity issues with trade policies. The resulting estimates

now support a wide class of trade and macro models in which large firms are less responsive to own cost shocks than small firms are.

Our findings on the micro-level firm heterogeneity in exposure and responses to input tariff shocks also have important welfare implications at the aggregate level. The fact that large firms benefit exclusively from the input tariff cuts yet do not fully pass through the shocks by raising their markups leads to increased markup dispersion and muted welfare gains. Neglecting the anti-competitive welfare effect can overestimate welfare gains from trade liberalization, during which we oftentimes see coincidence of output and input tariff cuts.

Our paper suggests that trade liberalization does not necessarily lower the aggregate price level or increase the welfare level by as much as policymakers may expect, especially when the big firms are exposed exclusively to input tariff cuts but not necessarily output tariff cuts if they are too good at lobbying. In such a case, the anti-competitive effects of trade reform can outweigh its pro-competitive effects, leading to a higher aggregate price and lower welfare level than in a counterfactual scenario with less unequal import access among domestic firms. Our work provides an example that policymakers in emerging economies should not blindly copy-paste successful reforms from developed countries given the disparities in pre-conditions. Our framework can also be extended to understand supply chain resilience at the micro level. While large firms are better able to diversify input sources, small firms are more likely to encounter supply chain bottlenecks as they largely source domestically. This calls for targeted intervention to enhance input access and substitutability especially for small business.<sup>28</sup>

While we focus on the first-order terms on firm-specific exposure and response to input tariff shocks, further questions can be asked about the second-order effects. Can we add an additional covariance term on a firm's market share and the composition of

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<sup>28</sup>DANE at Colombia also provides customs data that records firm-product-level import flows data at HS 10-digit. In addition, the dataset records the source country for each data entry. One could potentially merge the customs data with the EAM to study supply chain diversification at the firm level. We have not been able to merge the two due to their different firm coding systems.

its imported inputs? This can be especially relevant to political economics as it is to the interest of the large firms to lobby to have input tariffs lowered specifically for the products they themselves import intensively. Accounting for the endogeneity of tariff cuts should give even more significant anti-competitive effects of trade. Our focus of market power is restrained to the final goods market, and we have not opened the box of distortions in labor markets or intermediate product markets through the input tariff channel.<sup>29</sup> We leave related questions to future research.

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<sup>29</sup>See, for example, [Tortarolo and Zarate \(2018\)](#) and [Felix \(2021\)](#).

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# A Additional Figures and Tables

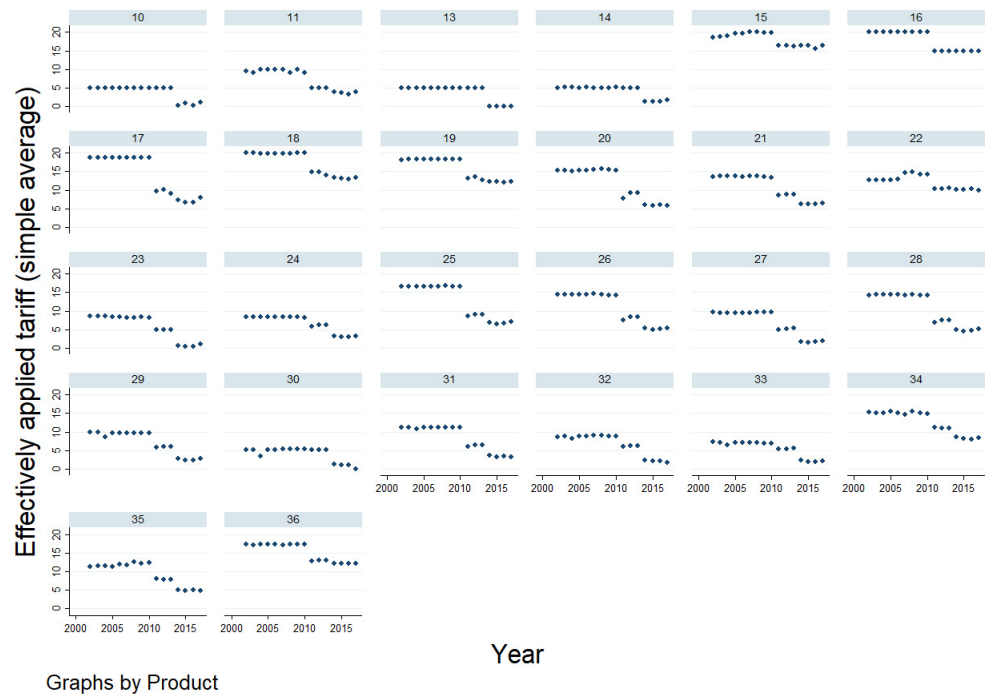


Figure A.1: Tariff Rates, Simple Mean, By Product, Percentage Points

Table A.1: Concentration Facts

|                               | Moments |                    | Moments |
|-------------------------------|---------|--------------------|---------|
| Within-sector concentration   |         |                    |         |
| Mean HHI                      | 0.30    | Mean inverse HHI   | 6.43    |
| Median HHI                    | 0.29    | Median inverse HHI | 3.25    |
| p10 HHI                       | 0.064   | p10 inverse HHI    | 1.58    |
| p25 HHI                       | 0.12    | p50 inverse HHI    | 2.89    |
| p75 HHI                       | 0.34    | p75 inverse HHI    | 7.83    |
| p90 HHI                       | 0.63    | p90 inverse HHI    | 15.08   |
| Distribution of market shares |         |                    |         |
| Mean share                    | 0.068   | Mean top share     | 0.44    |
| Median share                  | 0.00056 | Median top share   | 0.46    |
| p75 share                     | 0.020   | Mean CR4           | 0.75    |
| p90 share                     | 0.17    | Median CR4         | 0.84    |
| p95 share                     | 0.50    | Mean CR8           | 0.86    |
| SD share                      | 0.19    | Median CR8         | 0.95    |

Note: Summary statistics of main firm concentration facts using the final sample of EAM. The data uses the pre-shock firm distribution in year 2010. Producers' market shares are measured by their share of domestic sales revenue within a given 8-digit sector. HHI is the Herfindahl-Hirschman Index. The CR4 and CR8 are four-firm and eight-firm concentration ratios, defined as the market share of the 1st to the nth largest firm in an industry as a percentage of total industry market share.

Table A.2: Welfare Decomposition

| ISIC | Descriptions                                                   | OUT_NOCOMP | PRO_COMP | IN_NOCOMP | ANTI_COMP |
|------|----------------------------------------------------------------|------------|----------|-----------|-----------|
| 15   | Food products and beverages                                    | .01559     | .0096231 | .0995632  | -.0018021 |
| 16   | Tobacco products                                               | .01559     | .09      | .5482612  | 0         |
| 17   | Textiles                                                       | .0263048   | .0332848 | .284035   | -.0226498 |
| 18   | Wearing apparel; dressing and dyeing of fur                    | .0263048   | .0034724 | .0986912  | -.0032811 |
| 19   | Tanning and dressing of leather                                | .0263048   | .0123615 | .1359874  | -.005752  |
| 20   | Wood and of products of wood and cork, except furniture        | .0231405   | .0309218 | .0840705  | -.0157796 |
| 21   | Paper and paper products                                       | .0194271   | .0071859 | .2444555  | .0007039  |
| 22   | Publishing, printing and reproduction of recorded media        | .0194271   | .0036252 | .2697303  | -.0040856 |
| 23   | Coke, refined petroleum products and nuclear fuel              | .032766    | .0754328 | .3020252  | -.0423389 |
| 24   | Chemicals and chemical products                                | .0550142   | .0183806 | .3836652  | -.0162502 |
| 25   | Plastics products                                              | .0387577   | .0102586 | .3142702  | -.0010061 |
| 26   | Other non-metallic mineral products                            | .0162749   | .0067375 | .1434734  | .0002907  |
| 27   | Basic metals                                                   | .0553074   | .0129045 | .2152173  | .0036009  |
| 28   | Fabricated metal products, except machinery and equipment      | .0418605   | .0044067 | .2971714  | -.0096032 |
| 29   | Machinery and equipment n.e.c.                                 | .1460323   | .009064  | .226525   | -.0046542 |
| 30   | Office, accounting and computing machinery                     | .1961198   | .075773  | .6563208  | -.1211559 |
| 31   | Electrical machinery and apparatus n.e.c.                      | .0756014   | .0229864 | .2577966  | .0000909  |
| 32   | Radio, television and communication equipment and apparatus    | .1961198   | .0154723 | .3693828  | -.0087104 |
| 33   | Medical, precision and optical instruments, watches and clocks | .1961198   | .0110399 | .132975   | -.0043377 |
| 34   | Motor vehicles, trailers and semi-trailers                     | .0839094   | .0448139 | .379887   | -.0650204 |
| 35   | Other transport equipment                                      | .0980109   | .0433579 | .5898158  | -.0385084 |
| 36   | Furniture; manufacturing n.e.c.                                | .0559177   | .0085307 | .1369055  | .0134106  |

Note: Summary statistics of main firm concentration facts using the final sample of EAM. The data uses the pre-shock firm distribution in year 2010. Welfare effects are measured at 4-digit ISIC level and then aggregated to 2-digit level weighted by sales. OUT\_NOCOMP (IN\_NOCOMP) denotes the welfare effects of output (input) tariff liberalization without considering the competitive effects, and PRO\_COMP (ANTI\_COMP) the pro- (anti-)competitive effects considering markup distortions.

## B Proofs

### B.1 Proof of Proposition 1

Combining the price equations (6), (7), and (8) to get:

$$\begin{aligned}
d \ln P &= \sum_{i=1}^N s_i dp_i + s_f dp_f \\
&= \sum_{i=1}^N s_i dp_i + s_f d\tau_s \\
&= \sum_{i=1}^N s_i \left[ \frac{1-s_i}{1+(\sigma-2)s_i} dc_i + \frac{(\sigma-1)s_i}{1+(\sigma-2)s_i} d \ln P \right] + s_f d\tau_s \\
&= \sum_{i=1}^N s_i \left[ \frac{1-s_i}{1+(\sigma-2)s_i} \left( s_i^I d\tau_i^{input} + (1-s_i^I) d \ln w \right) + \frac{(\sigma-1)s_i}{1+(\sigma-2)s_i} d \ln P \right] + s_f d\tau_s \\
\implies d \ln P &= \sum_{i=1}^N \frac{s_i(1-s_i)}{1+(\sigma-2)s_i} \left[ (1-s_i^I) d \ln w + s_i^I d\tau_i^{input} \right] + \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} d \ln P + s_f d\tau_s
\end{aligned} \tag{12}$$

Rearranging terms, we obtain:

$$\begin{aligned}
&\sum_{i=1}^N \frac{s_i(1-s_i)}{1+(\sigma-2)s_i} \left[ (1-s_i^I) d \ln w + s_i^I d\tau_i^{input} \right] - \left[ 1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right] d \ln P + s_f d\tau_s = 0 \\
\implies &\sum_{i=1}^N \frac{s_i(1-s_i)}{1+(\sigma-2)s_i} \left( -s_i^I d \ln w + s_i^I d\tau_i^{input} \right) + \sum_{i=1}^N \frac{s_i(1-s_i)}{1+(\sigma-2)s_i} d \ln w \\
&\quad - \left[ 1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right] d \ln P + s_f d\tau_s = 0 \\
\implies &\sum_{i=1}^N \frac{s_i(1-s_i)}{1+(\sigma-2)s_i} \left( -s_i^I d \ln w + s_i^I d\tau_i^{input} \right) + \sum_{i=1}^N \frac{s_i(1-s_i)}{1+(\sigma-2)s_i} d \ln w \\
&\quad - \left[ 1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right] d \ln P + s_f d\tau_s \\
&\quad + \left[ 1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right] d \ln w - \left[ 1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right] d \ln w = 0
\end{aligned}$$

$$\begin{aligned}
&\Rightarrow \sum_{i=1}^N \frac{s_i(1-s_i)}{1+(\sigma-2)s_i} \left( -s_i^I d \ln w + s_i^I d \tau_i^{input} \right) + \left[ 1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right] (d \ln w - d \ln P) \\
&\quad + s_f d \tau_s + \left[ -d \ln w + \sum_{i=1}^N \frac{(\sigma-2)s_i^2}{1+(\sigma-2)s_i} d \ln w + \sum_{i=1}^N \frac{s_i}{1+(\sigma-2)s_i} d \ln w \right] = 0 \\
&\Rightarrow \sum_{i=1}^N \left[ \frac{(1-s_i)}{1+(\sigma-2)s_i} - \left[ 1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right] \right] \left[ s_i \left( s_i^I d \ln w - s_i^I d \tau_i^{input} \right) \right] \\
&\quad + \sum_{i=1}^N \left[ 1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right] \left[ s_i \left( s_i^I d \ln w - s_i^I d \tau_i^{input} \right) \right] \\
&\quad + \left[ d \ln w - \sum_{i=1}^N \frac{(\sigma-2)s_i^2}{1+(\sigma-2)s_i} d \ln w - \sum_{i=1}^N \frac{s_i}{1+(\sigma-2)s_i} d \ln w \right] = \\
&\quad \left[ 1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right] (d \ln w - d \ln P) + s_f d \tau_s \\
&\Rightarrow \sum_{i=1}^N \left[ 1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right] \left[ s_i \left( s_i^I d \ln w - s_i^I d \tau_i^{input} \right) \right] \\
&\quad - \left[ -d \ln w + \sum_{i=1}^N \frac{(\sigma-2)s_i^2}{1+(\sigma-2)s_i} d \ln w + \sum_{i=1}^N \frac{s_i}{1+(\sigma-2)s_i} d \ln w \right] = \\
&\quad \left[ 1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right] (d \ln w - d \ln P) + s_f d \tau_s \\
&\quad - \left[ 1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right] s_f (d \ln w - d \tau_s) + \left[ 1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right] s_f (d \ln w - d \tau_s) \\
&\quad - \sum_{i=1}^N \left[ \frac{(1-s_i)}{1+(\sigma-2)s_i} - \left[ 1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right] \right] \left[ s_i \left( s_i^I d \ln w - s_i^I d \tau_i^{input} \right) \right] \\
&\Rightarrow \left[ 1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right] \sum_{i=1}^N \left[ s_i \left( s_i^I d \ln w - s_i^I d \tau_i^{input} \right) \right] \\
&\quad + \left[ \left( 1 - \sum_{i=1}^N s_i \right) d \ln w \right] - s_f d \tau_s \\
&\quad + \left[ 1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right] s_f (d \ln w - d \tau_s) - \left[ 1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right] s_f (d \ln w - d \tau_s) \\
&\quad - \sum_{i=1}^N \left[ \left[ 1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right] - \frac{(1-s_i)}{1+(\sigma-2)s_i} \right] \left[ s_i \left( s_i^I d \ln w - s_i^I d \tau_i^{input} \right) \right] = \\
&\quad \left[ 1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right] (d \ln w - d \ln P)
\end{aligned}$$

$$\begin{aligned}
& \Rightarrow \left[ 1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right] \sum_{i=1}^N \left[ s_i \left( s_i^I d \ln w - s_i^I d \tau_i^{input} \right) \right] \\
& \quad + s_f d \ln w - s_f d \tau_s \\
& + \left[ 1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right] s_f (d \ln w - d \tau_s) - \left[ 1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right] s_f (d \ln w - d \tau_s) \\
& \quad - \sum_{i=1}^N \left[ \left[ 1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right] - \frac{(1-s_i)}{1+(\sigma-2)s_i} \right] \left[ s_i \left( s_i^I d \ln w - s_i^I d \tau_i^{input} \right) \right] \\
& \quad = \left[ 1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right] (d \ln w - d \ln P) \\
& \Rightarrow d \ln w - d \ln P = s_f (d \ln w - d \tau_s) + \sum_{i=1}^N \left[ s_i \left( s_i^I d \ln w - s_i^I d \tau_i^{input} \right) \right] \\
& \quad + s_f (d \ln w - d \tau_s) \frac{\sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}}{1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}} \\
& \quad - \sum_{i=1}^N \frac{\left[ \left[ 1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right] - \frac{(1-s_i)}{1+(\sigma-2)s_i} \right]}{1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}} s_i s_i^I (d \ln w - d \tau_i^{input}) \\
& \Rightarrow d \ln w - d \ln P = s_f (d \ln w - d \tau_s) + \sum_{i=1}^N \left[ s_i \left( s_i^I d \ln w - s_i^I d \tau_i^{input} \right) \right] \\
& \quad + s_f (d \ln w - d \tau_s) \frac{\sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}}{1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}} \\
& \quad - \frac{\sum_{i=1}^N \left[ \frac{(\sigma-1)s_i}{1+(\sigma-2)s_i} s_i (d \ln w - d \tau_i^{input}) \left( s_i^I - \sum_{i'=1}^N s_{i'}^I s_{i'} (d \ln w - d \tau_{i'}^{input}) \right) \right]}{1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}} \\
& \Rightarrow d \ln w - d \ln P = s_f [d \ln(w) - d \tau_s] + \sum_{i=1}^N \left[ s_i \left( s_i^I d \ln w - s_i^I d \tau_i^{input} \right) \right] \\
& \quad + s_f (d \ln w - d \tau_s) \frac{\sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}}{1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}} - \frac{Cov \left( s_i^I, \frac{(\sigma-1)s_i}{1+(\sigma-2)s_i}; s_i (d \ln w - d \tau_i^{input}) \right)}{1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}}
\end{aligned}$$

where the second last equality holds because

$$\begin{aligned}
& \frac{\sum_{i=1}^N \left[ \left[ 1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right] - \frac{(1-s_i)}{1+(\sigma-2)s_i} \right] s_i s_i^I \left( d \ln w - d\tau_i^{input} \right)}{1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}} \\
&= \frac{\sum_{i=1}^N \left[ 1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} - \frac{(1+(\sigma-2)s_i - (\sigma-1)s_i)}{1+(\sigma-2)s_i} \right] s_i s_i^I \left( d \ln w - d\tau_i^{input} \right)}{1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}} \\
&= \frac{\sum_{i=1}^N \left[ \frac{(\sigma-1)s_i}{1+(\sigma-2)s_i} s_i \left( d \ln w - d\tau_i^{input} \right) - s_i \left( d \ln w - d\tau_i^{input} \right) \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right] s_i^I}{1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}} \\
&= \frac{\sum_{i=1}^N \left[ \frac{(\sigma-1)s_i}{1+(\sigma-2)s_i} s_i \left( d \ln w - d\tau_i^{input} \right) s_i^I - s_i \left( d \ln w - d\tau_i^{input} \right) s_i^I \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right]}{1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}} \\
&= \frac{\sum_{i=1}^N \left[ \frac{(\sigma-1)s_i}{1+(\sigma-2)s_i} s_i \left( d \ln w - d\tau_i^{input} \right) s_i^I \right] - \sum_{i=1}^N s_i \left( d \ln w - d\tau_i^{input} \right) s_i^I \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}}{1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}} \\
&= \frac{\sum_{i=1}^N \left[ \frac{(\sigma-1)s_i}{1+(\sigma-2)s_i} s_i \left( d \ln w - d\tau_i^{input} \right) s_i^I \right] - \sum_{i=1}^N s_i \left( d \ln w - d\tau_i^{input} \right) s_i^I \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}}{1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}} \\
&= \frac{\sum_{i=1}^N \left[ \frac{(\sigma-1)s_i}{1+(\sigma-2)s_i} s_i \left( d \ln w - d\tau_i^{input} \right) s_i^I \right] - \sum_{i=1}^N \frac{(\sigma-1)s_i}{1+(\sigma-2)s_i} s_i \sum_{i=1}^N s_i \left( d \ln w - d\tau_i^{input} \right) s_i^I}{1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}} \\
&\Rightarrow \frac{\sum_{i=1}^N \left[ \left[ 1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i} \right] - \frac{(1-s_i)}{1+(\sigma-2)s_i} \right] s_i s_i^I}{1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}} \\
&= \frac{\sum_{i=1}^N \left[ \frac{(\sigma-1)s_i}{1+(\sigma-2)s_i} s_i \left( d \ln w - d\tau_i^{input} \right) \left( s_i^I - \sum_{i'=1}^N s_{i'} \left( d \ln w - d\tau_{i'}^{input} \right) s_{i'}^I \right) \right]}{1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}} \quad (13)
\end{aligned}$$

This allows us to solve for the following decomposition:

$$d\ln(w) - d\ln(P) =$$

$$\begin{aligned}
& \underbrace{s_f (d\ln w - d\tau_s) + \sum_{i=1}^N s_i s_i^I (d\ln(w) - d\tau_i^{input})}_{\text{Gains without competitive effects}} \\
& + \underbrace{s_f (d\ln w - d\tau_s) \frac{\sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}}{1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}}}_{\text{Pro-competitive effects on the final goods' market}} \\
& - \underbrace{\frac{Cov\left(s_i^I, \frac{(\sigma-1)s_i}{1+(\sigma-2)s_i}; s_i (d\ln w - d\tau_i^{input})\right)}{1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}}}_{\text{Anti-competitive effects of the input cost changes}}
\end{aligned}$$

where  $d\tau_{i,t}^{input} \equiv \sum_{s'} \theta_{i,s',t} d\log(1 + \tau_{s',t})$ .

## B.2 Generalization of Proposition 1

**Proposition 2.** *For any arbitrary invertible demand system and competition structure and under the assumption that i) the industry expenditure function is a sufficient statistic for competitor prices and ii) that the perceived demand elasticity is a function of the price of the firm relative to the industry expenditure function, the firm's markup change is the solution to the following fixed point equation:*

$$d\mu_{it} = \frac{\Gamma_{it}(1 - s_{it})}{1 - s_{it} + \Gamma_{it}} (-dmc_{it} + dmc_{-it}) + \frac{\Gamma_{it}}{1 - s_{it} + \Gamma_{it}} \underbrace{\left( \sum_{j=1}^N s_{jt} d\mu_{jt} \right)}_{\text{average markup change}}, \quad (14)$$

where  $dmc_{-it} \equiv \sum_{j \neq i}^N \frac{s_{jt}}{1 - s_{it}} dmc_{jt}$  is the index of competitor marginal cost changes.

*Proof.* Taking the price change decomposition equation from the [Amiti et al. \(2019\)](#) setup (assumptions from proposition 2, i) and rewriting the price changes in terms of markup changes using  $d\log(\mu_j) = d\log(p_j) - d\log(mc_j)$ , we obtain

$$\text{dlog}(\mu_i) = -\frac{\Gamma_i}{1 + \Gamma_i} \text{dlog}(mc_i) + \frac{\Gamma_{-i}}{1 + \Gamma_i} \sum_{j \neq i}^N \frac{s_j}{1 - s_i} (\text{dlog}(\mu_j) + \text{dlog}(mc_j)).$$

Let's collect all of the markup terms on the left-side to get

$$\begin{aligned} (1 + \Gamma_i)(1 - s_i) \text{dlog}(\mu_i) - \Gamma_{-i} \sum_{j \neq i}^N s_j \text{dlog}(\mu_j) &= -\Gamma_i(1 - s_i) \text{dlog}(mc_i) + \\ &+ \Gamma_{-i} \sum_{j \neq i}^N s_j \text{dlog}(mc_j). \end{aligned}$$

This could be interpreted as an equation for firm markup change decomposition

$$\begin{aligned} \text{dlog}(\mu_i) &= \frac{\Gamma_{-i}}{1 - s_i + \Gamma_i} \left( \frac{\Gamma_i}{\Gamma_{-i}} s_i \text{dlog}(\mu_i) + \sum_{j \neq i}^N s_j \text{dlog}(\mu_j) \right) \\ &- \frac{\Gamma_i}{1 - s_i + \Gamma_i} (1 - s_i) \text{dlog}(mc_i) + \frac{\Gamma_{-i}}{1 - s_i + \Gamma_i} \sum_{j \neq i}^N s_j \text{dlog}(mc_j). \end{aligned}$$

If  $\Gamma_i = \Gamma_{-i}$  (assumptions from proposition 2, ii in [Amiti et al. \(2019\)](#)), then the whole expression simplifies to

$$\begin{aligned} \text{dlog}(\mu_i) &= \frac{\Gamma_i}{1 - s_i + \Gamma_i} \underbrace{\left( \sum_{j=1}^N s_j \text{dlog}(\mu_j) \right)}_{\text{average (weighted) markup change}} \\ &+ \frac{\Gamma_i}{1 - s_i + \Gamma_i} \left[ -(1 - s_i) \text{dlog}(mc_i) + \sum_{j \neq i}^N s_j \text{dlog}(mc_j) \right]. \end{aligned}$$

□

**Corollary 2.** If the marginal cost changes by the same percentage for all firms, i.e.,  $\text{dmc}_i = \text{dmc}_j \forall i, j$ , we have

$$d\mu_i = 0 \forall i.$$

The proof immediately follows from Proposition 2.

**Proposition 3.** *For any arbitrary invertible demand system and competition structure and under the assumption that i) the industry expenditure function is a sufficient statistic for competitor prices and ii) that the perceived demand elasticity is a function of the price of the firm relative to the industry expenditure function, the change in the price index is given by:*

$$dp = \underbrace{s_f d\tau + \sum_{i=1}^N s_i z_i d\tau_i^{input}}_{\text{Gains without competitive effects}} + \underbrace{s_f d\tau \frac{\bar{\kappa}}{1 - \bar{\kappa}}}_{\text{Pro-competitive effect}} - \underbrace{\frac{\text{cov}(\kappa_i, z_i d\tau_i^{input}; s_i)}{1 - \bar{\kappa}}}_{\text{Anti-competitive effect}},$$

where  $\kappa_i \equiv \frac{\Gamma_i}{1 - s_i + \Gamma_i}$ ,  $\bar{\kappa} \equiv \sum_{i=1}^N s_i \kappa_i$ , and  $\text{cov}(a_i, b_i; c_i)$  is the weighted covariance between  $a_i$  and  $b_i$  with weights  $c_i$ .

*Proof.* We use the fix point equation (14) and add over all  $i$ 's to obtain:

$$\begin{aligned}
d\bar{\mu}_t &= \sum_{i=1}^N s_{it} d\mu_{it} = \sum_{i=1}^N s_{it} \left[ \frac{\Gamma_{it}(1-s_{it})}{1-s_{it}+\Gamma_{it}} (-dmc_{it} + dmc_{-it}) + \frac{\Gamma_{it}}{1-s_{it}+\Gamma_{it}} d\bar{\mu}_t \right] \\
d\bar{\mu}_t &= \sum_{i=1}^N s_{it} [-\kappa_{it}(1-s_{it}) (dmc_{it} - dmc_{-it}) + \kappa_{it} d\bar{\mu}_t] \\
(1-\bar{\kappa}_t) d\bar{\mu}_t &= - \sum_{i=1}^N s_{it} [\kappa_{it}(1-s_{it}) (dmc_{it} - dmc_{-it})] \\
&= - \sum_{i=1}^N s_{it} \left[ \kappa_{it}(1-s_{it}) \left( dmc_{it} - \sum_{j \neq i}^N \frac{s_{jt}}{1-s_{it}} dmc_{jt} \right) \right] \\
&= - \sum_{i=1}^N s_{it} \left[ \kappa_{it}(1-s_{it}) dmc_{it} - \kappa_{it} \sum_{j \neq i}^N s_{jt} dmc_{jt} \right] \\
&= - \sum_{i=1}^N s_{it} \left[ \kappa_{it} dmc_{it} - \kappa_{it} \sum_j s_{jt} dmc_{jt} \right] \\
(1-\bar{\kappa}_t) d\bar{\mu}_t &= - \sum_{i=1}^N s_{it} \kappa_{it} [dmc_{it} - d\bar{mc}_t] \\
(1-\bar{\kappa}_t) d\bar{\mu}_t &= -\mathbf{cov}(\kappa_{it}, dmc_{it}; s_{it})
\end{aligned}$$

Now combine with the average change in marginal cost  $d\bar{mc} = s_f d\tau + \sum_{i=1}^N s_i z_i d\tau_i^{input}$ , we obtain

$$\begin{aligned}
dp &= d\bar{mc} + d\bar{\mu} = s_f d\tau + \sum_{i=1}^N s_i z_i d\tau_i^{input} - \sum_{i=1}^N s_i \frac{\kappa_i}{1-\bar{\kappa}} [dmc_i - d\bar{mc}] \\
&= s_f d\tau + \sum_{i=1}^N s_i z_i d\tau_i^{input} - \sum_{i=1}^N \frac{\kappa_i}{1-\bar{\kappa}} s_i \left[ z_i d\tau_i^{input} - \sum_{i=1}^N s_i z_i d\tau_i^{input} \right] + s_f d\tau \sum_{i=1}^N \frac{s_i \kappa_i}{1-\bar{\kappa}_t} \\
&= s_f d\tau + \sum_{i=1}^N s_i z_i d\tau_i^{input} - \frac{\sum_{i=1}^N s_i \kappa_i [z_i d\tau_i^{input} - \sum_{i=1}^N s_i z_i d\tau_i^{input}]}{1-\bar{\kappa}} + s_f d\tau \frac{\bar{\kappa}}{1-\bar{\kappa}} \\
\Rightarrow dp &= \underbrace{s_f d\tau + \sum_{i=1}^N s_i z_i d\tau_i^{input}}_{\text{Gains without competitive effects}} + \underbrace{s_f d\tau \frac{\bar{\kappa}}{1-\bar{\kappa}}}_{\text{Pro-competitive effect}} - \underbrace{\frac{\mathbf{cov}(\kappa_i, z_i d\tau_i^{input}; s_i)}{1-\bar{\kappa}}}_{\text{Anti-competitive effect}}
\end{aligned}$$

Note that this is a general version of Proposition 1, which is a special case when  $\kappa_{it} = \frac{(\sigma-1)s_i}{1+(\sigma-2)s_i}$  :

$$\begin{aligned}
dp = & \underbrace{s_f d\tau + \sum_{i=1}^N s_i z_i d\tau_i^{input}}_{\text{Gains without competitive effects}} + \underbrace{s_f d\tau \frac{\sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}}{1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}}}_{\text{Pro-competitive effect}} \\
& + \underbrace{\frac{\text{cov}\left(z_i, \frac{(\sigma-1)s_i}{1+(\sigma-2)s_i}; s_i \left(-d\tau_i^{input}\right)\right)}{1 - \sum_{i=1}^N \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}}}_{\text{Anti-competitive effect}}
\end{aligned}$$

□

## C Simulations

### C.1 Model for Simulation (Single Importer)

#### Consumers

Under the Atkeson-Burstein setting, we have Cournot oligopoly for the final sector with CES elasticity of substitution parameter  $\sigma$ . Consider a representative consumer household that supplies one unit of labor and earns national income ( $I$ ). Consumer preferences are given by the following utility function:

$$U = \left[ \sum_{i=1}^n \beta_i q_i^{\frac{\sigma-1}{\sigma}} + \beta_f q_f^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}} = \frac{I}{\left[ \sum_{i=1}^n \beta_i^\sigma p_i^{1-\sigma} + \beta_f^\sigma p_f^{1-\sigma} \right]^{\frac{1}{1-\sigma}}} = \frac{I}{P}$$

There are  $n$  final goods producers in the economy. Demand parameter for firm  $i$  is equal to  $\beta_i$ , while  $\beta_f$  is the parameter guiding consumer demand for the foreign goods.

Additionally, there is demand for domestic firms' final products on the foreign market:

$$q_{if} = \beta_{if}^\sigma p_{if}^{-\sigma}$$

## Firms

Domestic firms that produce final goods combine labor and intermediate inputs in their production. There is one domestic intermediate producer which prices at CES markups and uses only labor. Intermediate good producer does not sell to consumers. Intermediate inputs are tradable and produced competitively or with constant markups using labor. Each firm  $i$  produces using a Cobb-Douglas production function, with CES aggregate of intermediate goods either domestically sourced or imported. This leads to the following cost function:

$$c_i = \frac{w^{\alpha_i} (c_i^m)^{1-\alpha_i}}{\phi_i}$$

where

$$c_i^m = \left( \alpha_{di}^\rho (p_d^m)^{1-\rho} + (P_{fi}^m)^{1-\rho} \right)^{\frac{1}{1-\rho}}$$

with

$$P_{fi}^m = \left( \sum_s \alpha_{fsi}^\rho (p_{fs}^m)^{1-\rho} \right)^{\frac{1}{1-\rho}}$$

Importing intermediate inputs incurs trade costs  $\tau_s$ . Foreign firms producing final goods have marginal costs equal to  $c_f$ , and foreign firms producing intermediate goods have marginal costs equal to  $c_{fs}^m$ . Assume that foreign firms do not take part in the cournot competition with domestic firms, thus fully passing through the cost shocks.

## Equilibrium

Let's first write down cost and income as functions of the vector of all markups in the economy.<sup>30</sup> The markup vector consists of  $n$  markups of domestic firms on final goods market ( $\mu_i$ ),  $n$  markups of domestic firms on intermediate goods market ( $\mu_{mi}$ ),  $n$  markups of domestic firms on foreign market ( $\mu_{if}$ ), and  $n + 1$  foreign markups ( $\mu_f$ , and  $\mu_{fi}$ ). Under Cournot competition in the final goods market, firm  $i$ 's markup  $\mu_i$  depends on its market share:

$$\mu_i = \frac{\epsilon_i}{\epsilon_i - 1}$$

$$\text{where } \epsilon_i = \left[ s_i + \frac{1}{\sigma} (1 - s_i) \right]^{-1}.$$

Below are expressions for marginal cost of firm  $i$ , its total output and sales:

$$c_i = \frac{w^{\alpha_i}}{\phi_i} \left[ \alpha_{di}^{\rho} (\mu_i^m w)^{1-\rho} + \sum_s \alpha_{fsi}^{\rho} (\tau_s \mu_{fs} c_{fs}^m)^{1-\rho} \right]^{\frac{1-\alpha_i}{1-\rho}}$$

$$q_i = \mu_i^{-\sigma} \beta_i^{\sigma} c_i^{-\sigma} \frac{I}{p^{1-\sigma}} + \mu_{if}^{-\sigma} \beta_{if}^{\sigma} c_i^{-\sigma}$$

$$\pi_i = \frac{\mu_i - 1}{\mu_i^{\sigma}} \beta_i^{\sigma} c_i^{1-\sigma} \frac{I}{p^{1-\sigma}} + \frac{\mu_{if} - 1}{\mu_{if}^{\sigma}} \beta_{if}^{\sigma} c_i^{1-\sigma}$$

$$s_i = \mu_i^{1-\sigma} \beta_i^{\sigma} c_i^{1-\sigma} \frac{I}{p^{1-\sigma}}$$

For the intermediate firm:

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<sup>30</sup>We later allow for alternative markup equation to determine markups in the equilibrium: for example, 1) constant markups (Melitz 2003, ACR 2012, Tintelnot et al. 2020), 2) markups depending on elasticities and market shares (Atkeson and Burstein 2008, Kikkawa et al. 2019), 3) non-CES monopolistic competition (ACDR 2019), 4) markups depending on Lagrange multiplier for additive VES preferences (Krugman 1979, Dhingra and Morrow 2019), etc.

$$\pi_m = \sum_{i=1}^n \frac{1}{\eta} \left( \frac{1 - \alpha_i}{1 + \mu_{fi}^{1-\rho} \alpha_{fi}^\rho (\tau_i^{in} c_{mf} / w)^{1-\rho}} \right) \left[ \beta_i^\sigma \mu_i^{-\sigma} c_i^{1-\sigma} \frac{I}{p^{1-\sigma}} + \beta_{if}^\sigma \mu_{if}^{-\sigma} c_i^{1-\sigma} \right],$$

where  $\eta$  is the CES elasticity of substitution parameter for the intermediate sector.

**Closing the model:** The two conditions that pin down the equilibrium of the model are: 1) expression for total domestic income incorporating the labor market clearing condition:

$$\sum_{i=1}^n \pi_i + \pi_m + w = I,$$

and 2) trade balance equation:

$$\sum_{i=1}^n \mu_{if}^{1-\sigma} \beta_{if}^\sigma c_i^{1-\sigma} = \sum_{i=1}^n \frac{\mu_{fi}^{1-\rho} \alpha_{fi}^\rho (\tau_i^{in} c_{mf})^{1-\rho}}{(p_{mi})^{1-\rho}} c_i q_i + \beta_f^\sigma (p_f)^{1-\sigma} \frac{I}{p^{1-\sigma}}.$$

### Parameter Restrictions

There is 1 large firms in the economy, 9 small ones, and 1 intermediate good producer let  $n = 1 + 9 + 1 = 11$ .

For large firm demand parameter is equal to  $\beta_1 = 5$  and  $\beta_{1F} = 5$ , and for small firms demand parameter is equal to  $\beta_i = 1$  and  $\beta_{iF} = 0$ ,  $\beta_F = 5$  is the parameter guiding consumer demand for the foreign goods. Intermediate good producer does not sell to consumers. Foreign cost is  $c_F = 1$ , and  $\tau = 1.25$ ,  $\tau' = 1$ .

Labor shares:  $\alpha_1 = 1$ ,  $\alpha_i = 2/3$  for  $2 \leq i \leq 10$ . Intermediate cost parameters:  $\phi_i = 1$  for everybody,  $\alpha_{1i} = 1$  for everybody,  $\alpha_{F1} = 1$  and  $\alpha_{Fi} = 0$  for  $2 \leq i \leq 10$ ,  $c_F^m = 1$ .

Assume  $\sigma = 5$ ,  $\rho = 2$ . For markups: assume  $\mu_i = \mu_{iF} = \mu_F = \sigma/(\sigma - 1)$  for  $2 \leq i \leq 10$ ,  $\mu_1 = 1.5$ ,  $\mu_{ji} = \rho/(\rho - 1)$ .

## System of Equations

Below is the simplified version with only 1 intermediate good producer which uses only labor:

$$c_i = \frac{w^{\alpha_i}}{\phi_i} \left[ \mu_{ji}^{1-\rho} \alpha_{ji}^\rho w^{1-\rho} + \mu_{Fi}^{1-\rho} \tau^{1-\rho} \alpha_{Fi}^\rho (c_F^m)^{1-\rho} \right]^{\frac{1-\alpha_i}{1-\rho}}$$

$$P = \left[ \sum_{i=1}^n \beta_i^\sigma p_i^{1-\sigma} + \beta_F^\sigma p_F^{1-\sigma} \right]^{\frac{1}{1-\sigma}}$$

$$p_{mi} = \left( \alpha_i^\rho (\mu_{ji} w)^{1-\rho} + \alpha_{Fi}^\rho p_{Fi}^{1-\rho} \right)^{\frac{1}{1-\rho}}$$

$$q_i = \mu_i^{-\sigma} \beta_i^\sigma c_i^{-\sigma} \frac{I}{p^{1-\sigma}} + \mu_{iF}^{-\sigma} \beta_{iF}^\sigma c_i^{-\sigma}$$

$$q_i = \mu_i^{-\sigma} \beta_i^\sigma c_i^{-\sigma} \frac{I}{p^{1-\sigma}} + \mu_{iF}^{-\sigma} \beta_{iF}^\sigma c_i^{-\sigma} + \sum_{j \neq i}^n \frac{\mu_{ij}^{-\rho} \alpha_{ij}^\rho c_i^{-\rho}}{p_{mj}^{1-\rho}} c_j q_j$$

$$\pi_i = \frac{\mu_i - 1}{\mu_i^\sigma} \beta_i^\sigma c_i^{1-\sigma} \frac{I}{p^{1-\sigma}} + \frac{\mu_{iF} - 1}{\mu_{iF}^\sigma} \beta_{iF}^\sigma c_i^{1-\sigma} + \sum_{j \neq i}^n \frac{\mu_{ij} - 1}{\mu_{ij}^\rho} \frac{\alpha_{ij}^\rho c_i^{1-\rho}}{p_{mj}^{1-\rho}} c_j q_j$$

$$\begin{aligned}
I &= \sum_{i=1}^n \pi_i + \sum_{i=1}^n \alpha_i c_i q_i \\
&= \sum_{i=1}^n \left( \frac{\mu_i - 1}{\mu_i^\sigma} \beta_i^\sigma c_i^{1-\sigma} \frac{I}{p^{1-\sigma}} + \frac{\mu_{iF} - 1}{\mu_{iF}^\sigma} \beta_{iF}^\sigma c_i^{1-\sigma} + \sum_{j \neq i} \frac{\mu_{ij} - 1}{\mu_{ij}^\rho} \frac{\alpha_{ij}^\rho c_i^{1-\rho}}{p_{mj}^{1-\rho}} c_j q_j \right) \\
&\quad + \sum_{i=1}^n \left( \alpha_i \mu_i^{-\sigma} \beta_i^\sigma c_i^{1-\sigma} \frac{I}{p^{1-\sigma}} + \alpha_i \mu_{iF}^{-\sigma} \beta_{iF}^\sigma c_i^{1-\sigma} + \sum_{j \neq i} \frac{\alpha_i \mu_{ij}^{-\rho} \alpha_{ij}^\rho c_i^{1-\rho}}{p_{mj}^{1-\rho}} c_j q_j \right) \\
&\Rightarrow I - \sum_{i=1}^n \left( \frac{\mu_i - 1}{\mu_i^\sigma} + \alpha_i \mu_i^{-\sigma} \right) \beta_i^\sigma c_i^{1-\sigma} \frac{I}{p^{1-\sigma}} \\
&= \sum_{i=1}^n \left[ \left( \frac{\mu_{iF} - 1}{\mu_{iF}^\sigma} + \alpha_i \mu_{iF}^{-\sigma} \right) \beta_{iF}^\sigma c_i^{1-\sigma} + \sum_{j \neq i} \left( \frac{\mu_{ij} - 1}{\mu_{ij}^\rho} + \alpha_i \mu_{ij}^{-\rho} \right) \frac{\alpha_{ij}^\rho c_i^{1-\rho}}{p_{mj}^{1-\rho}} c_j q_j \right] \\
&\Rightarrow I = \frac{\sum_{i=1}^n \left[ \left( \frac{\mu_{iF} - 1}{\mu_{iF}^\sigma} + \alpha_i \mu_{iF}^{-\sigma} \right) \beta_{iF}^\sigma c_i^{1-\sigma} + \sum_{j \neq i} \left( \frac{\mu_{ij} - 1}{\mu_{ij}^\rho} + \alpha_i \mu_{ij}^{-\rho} \right) \frac{\alpha_{ij}^\rho c_i^{1-\rho}}{p_{mj}^{1-\rho}} c_j q_j \right]}{1 - \sum_{i=1}^n \left( \frac{\mu_i - 1}{\mu_i^\sigma} + \alpha_i \mu_i^{-\sigma} \right) \beta_i^\sigma c_i^{1-\sigma} \frac{1}{p^{1-\sigma}}}
\end{aligned}$$

$$\sum_{i=1}^n \mu_{iF}^{1-\sigma} \beta_{iF}^\sigma c_i^{1-\sigma} = \sum_{i=1}^n \frac{\mu_{Fi}^{1-\rho} \alpha_{Fi}^\rho \tau^{1-\rho} (c_F^m)^{1-\rho}}{p_{mi}^{1-\rho}} c_i q_i + \mu_F^{1-\sigma} \beta_F^\sigma \tau^{1-\sigma} (c_F^m)^{1-\sigma} \frac{I}{p^{1-\sigma}}$$

Things we assume throughout --- there is one large firm (indexed 1),  $n$  small ones (indexed 2 to  $n+1$ ), and intermediate producer (indexed  $n+2$ ), small firms do not export or import,  $\beta_i$  for small firms is 1, small and intermediate domestic firms price at CES markups.

We need first to write down expression for income as a function of wage and parameters:

$$I = \sum_{i=1}^{n+2} \pi_i + w$$

Let's denote labor share of firms producing final goods simply as  $\alpha$ . Given that small firms do not use imported intermediate inputs (we can model this as either

$\alpha_{Fi} = 0$  for  $2 \leq i \leq n+1$  or by assuming small has additional costs of importing so for them  $\tau_i = \tau \times d_i$  with  $d_i \rightarrow \infty$ ),  $c_{11} = w$  and other productivity parameters:  $\phi_i = 1$  and normalizing  $\alpha_{11i} = 1$  for  $1 \leq i \leq n+1$ , we have for  $2 \leq i \leq n+1$ :

$$c_i = \frac{w}{\phi_i} (\mu_{ji})^{1-\alpha} = w \left( \frac{\rho}{\rho-1} \right)^{1-\alpha}$$

The cost of the large firm assuming  $\phi_1 = 1$  and  $\alpha_{F1} = \alpha_F$  is given by:

$$c_1 = w \left[ \left( \frac{\rho}{\rho-1} \right)^{1-\rho} + \tau^{1-\rho} \alpha_F^\rho \left( \frac{p_F^m}{w} \right)^{1-\rho} \right]^{\frac{1-\alpha}{1-\rho}} = c_i \left[ 1 + \tau^{1-\rho} \alpha_F^\rho \left( \frac{(\rho-1)p_F^m}{\rho w} \right)^{1-\rho} \right]^{\frac{1-\alpha}{1-\rho}}$$

$$c_1/c_i = \left[ 1 + \tau^{1-\rho} \alpha_F^\rho \left( \frac{(\rho-1)p_F^m}{\rho w} \right)^{1-\rho} \right]^{\frac{1-\alpha}{1-\rho}}$$

This gives expression for price index as function of wage and given parameters:

$$P = \left[ w^{1-\sigma} \underbrace{\left( \frac{\rho}{\rho-1} \right)^{(1-\alpha)(1-\sigma)}}_{\psi} \left( \beta_1^\sigma \mu_1^{1-\sigma} \left( \frac{c_1}{c_i} \right)^{1-\sigma} + n \left( \frac{\sigma}{\sigma-1} \right)^{1-\sigma} \right) + \beta_F^\sigma (\tau p_F)^{1-\sigma} \right]^{\frac{1}{1-\sigma}}$$

Let's write down expression for profits. First, for small firms given  $\beta_i = 1$ :

$$\pi_i = \psi \frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} \frac{I}{P^{1-\sigma}} w^{1-\sigma}$$

For the large firm:

$$\pi_1 = \psi \left( \frac{c_1}{c_i} \right)^{1-\sigma} w^{1-\sigma} \left( \frac{\mu_1 - 1}{\mu_1^\sigma} \beta_1^\sigma \frac{I}{P^{1-\sigma}} + \frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} \beta_{1F}^\sigma \right)$$

For the intermediate firm we need to solve for the share that large firm spends on

domestic intermediate goods:

$$s_{11} = \frac{1 - \alpha}{1 + \tau^{1-\rho} \alpha_F^\rho \left( \frac{(\rho-1)p_F^m}{\rho w} \right)^{1-\rho}} = \frac{1 - \alpha}{1 + (c_1/c_i)^{\frac{1-\rho}{1-\alpha}} - 1} = (1 - \alpha)(c_1/c_i)^{\frac{\rho-1}{1-\alpha}}$$

Share spent on imported inputs is:

$$s_I = (1 - \alpha) \left( 1 - (c_1/c_i)^{\frac{\rho-1}{1-\alpha}} \right)$$

Then profits of the intermediate firm are given by:

$$\pi_{n+2} = \frac{(1 - \alpha)}{\rho} \left[ \psi \left( \frac{c_1}{c_i} \right)^{\frac{\rho-\alpha}{1-\alpha}-\sigma} \left( \frac{\beta_1^\sigma}{\mu_1^\sigma} \frac{I}{P^{1-\sigma}} + \frac{(\sigma-1)^\sigma}{\sigma^\sigma} \beta_{1F}^\sigma \right) w^{1-\sigma} + n\pi_2(\sigma-1) \right]$$

Expression for income is then a function of wage:

$$I = \pi_1 + n\pi_2 + \pi_{n+2} + w \Rightarrow$$

$$\begin{aligned} I &= n\psi \frac{Iw^{1-\sigma}}{P^{1-\sigma}} \frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} \left( 1 + \frac{(1-\alpha)(\sigma-1)}{\rho} \right) + \\ &+ \psi \frac{Iw^{1-\sigma}}{P^{1-\sigma}} \left( \frac{c_1}{c_i} \right)^{1-\sigma} \left( \frac{\beta_1}{\mu_1} \right)^\sigma \left[ \frac{(1-\alpha)}{\rho} \left( \frac{c_1}{c_i} \right)^{\frac{\rho-1}{1-\alpha}} + \mu_1 - 1 \right] + \\ &+ \psi \left( \frac{c_1}{c_i} \right)^{1-\sigma} w^{1-\sigma} \frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} \beta_{1F}^\sigma \left[ \frac{(1-\alpha)(\sigma-1)}{\rho} \left( \frac{c_1}{c_i} \right)^{\frac{\rho-1}{1-\alpha}} + 1 \right] + w \Rightarrow \\ &w + \psi \left( \frac{c_1}{c_i} \right)^{1-\sigma} w^{1-\sigma} \frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} \beta_{1F}^\sigma \left[ \frac{(1-\alpha)(\sigma-1)}{\rho} \left( \frac{c_1}{c_i} \right)^{\frac{\rho-1}{1-\alpha}} + 1 \right] \\ I &= \frac{w + \psi \left( \frac{c_1}{c_i} \right)^{1-\sigma} w^{1-\sigma} \frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} \beta_{1F}^\sigma \left[ \frac{(1-\alpha)(\sigma-1)}{\rho} \left( \frac{c_1}{c_i} \right)^{\frac{\rho-1}{1-\alpha}} + 1 \right]}{1 - \psi \frac{w^{1-\sigma}}{P^{1-\sigma}} \left[ n \frac{\rho + (1-\alpha)(\sigma-1)}{\rho} \frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} + \left( \frac{c_1}{c_i} \right)^{1-\sigma} \left( \frac{\beta_1}{\mu_1} \right)^\sigma \left[ \frac{(1-\alpha)}{\rho} \left( \frac{c_1}{c_i} \right)^{\frac{\rho-1}{1-\alpha}} + \mu_1 - 1 \right] \right]} \end{aligned}$$

Next, use trade balance equation.

## C.2 Algorithm for Simulation (Single Importer)

1. Set number of small firms  $n$ .
2. Set parameter values (  $\underbrace{\alpha, \sigma, \rho, \alpha_{n+2}, \alpha_F}_{\text{elasticities, cost shares}} \quad \underbrace{\gamma, \beta_1, \beta_{1F}}_{\text{demand for large firm}} \quad \underbrace{\tau, p_F^m, p_F, \beta_F}_{\text{importer parameters}}$  )
3. Assume value for  $w$ .
4. Preliminary expressions:

$$\psi = \left( \frac{\rho}{\rho - 1} \right)^{(1-\alpha)(1-\sigma)}$$

$$c_1/c_i = \left[ 1 + \tau^{1-\rho} \alpha_F^\rho \left( \frac{(\rho - 1)p_F^m}{\rho w} \right)^{1-\rho} \right]^{\frac{1-\alpha}{1-\rho}}$$

$$s_I = (1 - \alpha) \left( 1 - (c_1/c_i)^{\frac{\rho-1}{1-\alpha}} \right)$$

5. Solving for  $\mu_1$ .

Given guess for  $\mu_1$ , we have:

$$P = \left[ \psi w^{1-\sigma} \left( \beta_1^\sigma \mu_1^{1-\sigma} \left( \frac{c_1}{c_i} \right)^{1-\sigma} + n \left( \frac{\sigma}{\sigma - 1} \right)^{1-\sigma} \right) + \tau^{1-\sigma} \beta_F^\sigma p_F^{1-\sigma} \right]^{\frac{1}{1-\sigma}}$$

Error from markup equation:

$$\mu_1 - \gamma \mu_1^{1-\sigma} \frac{\psi (\beta_1)^\sigma (c_1/c_i)^{1-\sigma} w^{1-\sigma}}{P^{1-\sigma}} = \frac{\sigma}{\sigma - 1}$$

6. Iterate  $\mu_1$  until error  $\approx 0$ .

7. Expression for income:

$$I = \frac{w + \psi \left( \frac{c_1}{c_i} \right)^{1-\sigma} w^{1-\sigma} \frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} \beta_{1F}^\sigma \left[ \frac{(1-\alpha)(\sigma-1)}{\rho} \left( \frac{c_1}{c_i} \right)^{\frac{\rho-1}{1-\alpha}} + 1 \right]}{1 - \psi \frac{w^{1-\sigma}}{p^{1-\sigma}} \left[ n \frac{\rho+(1-\alpha)(\sigma-1)}{\rho} \frac{(\sigma-1)^{\sigma-1}}{\sigma^\sigma} + \left( \frac{c_1}{c_i} \right)^{1-\sigma} \left( \frac{\beta_1}{\mu_1} \right)^\sigma \left[ \frac{1-\alpha}{\rho} \left( \frac{c_1}{c_i} \right)^{\frac{\rho-1}{1-\alpha}} + \mu_1 - 1 \right] \right]}$$

8. Error from trade balance:

$$\underbrace{\psi \beta_{1F}^\sigma \left( \frac{c_1}{c_i} \right)^{1-\sigma} \left( \frac{\sigma}{\sigma-1} \right)^{1-\sigma} w^{1-\sigma}}_{\text{exports}} - \underbrace{\left( s_I \psi \left( \frac{c_1}{c_i} \right)^{1-\sigma} \left( \frac{\beta_1^\sigma}{\mu_1^\sigma} \frac{I}{p^{1-\sigma}} + \frac{(\sigma-1)^\sigma}{\sigma^\sigma} \beta_{1F}^\sigma \right) w^{1-\sigma} + \tau^{1-\sigma} \beta_F^\sigma (p_F)^{1-\sigma} \frac{I}{p^{1-\sigma}} \right)}_{\text{imports}}$$

9. Iterate  $w$  until error  $\approx 0$ .

### C.3 Model for Simulation (Many Importers)

**Firms and workers:** There are  $N$  domestic firms (indexed  $i$ ) and a foreign exporter (indexed  $F$ ), intermediate domestic producer (indexed  $ID$ ) and intermediate exporter (indexed  $IF$ ), intermediate domestic producer prices at CES markups and uses only labor. Domestic firms in the final sector engage in Cournot competition a-la Atkeson-Burstein (2008). Exporters are for now assumed to price competitively (equivalently to fixed markups). Total labor force is 1 and is supplied inelastically. We normalize everything relative to fixed foreign wage  $w_F = 1$ .

**Parameters:** Each domestic firm is characterised by a set of 4 different (potentially correlated) parameters:  $(\beta_i, \beta_{Fi}, \alpha_{Fi}, \phi_i)$  --- shifters of domestic demand, foreign demand, importing share and efficiency. Demand for foreign goods in the final sector

is determined by parameter  $\beta_F$ . For domestic intermediate producer productivity parameter  $\phi_{ID} = 1$  and foreign demand is given by parameter  $\beta_{FID}$ . Importing trade costs for the final sector are  $(\tau_F, \tau_I)$ . Labor share of firms producing final goods is  $\alpha$ . E-o-s in the final sector is  $\sigma$ , e-o-s in the intermediate sector is  $\rho$ .

**Solution:** We want to solve for two general equilibrium objects  $(I, P)$  --- domestic nominal income and final sector price index --- in terms of domestic wage. Then we can write down trade balance equation to pin down the domestic wage. First, write down expression for income as a function of wage and parameters:

$$I = \sum_{i=1}^N \pi_i + \pi_{ID} + w \quad (15)$$

To do that we need to solve for firms' costs to then derive their profits. To simplify the notation, let's normalize the production function in the following way:

$$q_i = \left( \frac{\rho^{1-\alpha}}{\alpha^\alpha ((1-\alpha)(\rho-1))^{1-\alpha}} \right) l_i^\alpha m_i^{1-\alpha} \quad (16)$$

$$m_i = \left( m_{Di}^{\frac{\rho-1}{\rho}} + \alpha_{Fi} \left( \frac{\rho-1}{\rho} \right)^{\frac{\rho-1}{\rho}} m_{Fi}^{\frac{\rho-1}{\rho}} \right)^{\frac{\rho}{\rho-1}} \quad (17)$$

This is equivalent from the perspective of firm to assuming that there is firm-specific importing iceberg cost  $d_i = \frac{\rho}{\rho-1} \alpha_{Fi}^{-\frac{\rho}{\rho-1}}$ .

Cost minimization gives us the marginal cost of the firm  $i$  as a function of  $w$ :

$$c_i(w) = \frac{w}{\phi_i} \left[ 1 + \tau_I^{1-\rho} \alpha_{Fi}^\rho w^{\rho-1} \right]^{\frac{1-\alpha}{1-\rho}} \quad (18)$$

Let's now derive parts of profit functions that depend on exogenous markups --- export profits of intermediate producer and export profits of final producers.

Given the firm-specific foreign demand function for final goods, we derive export profit of the final producers:

$$q_{Fi} = \left( \frac{\sigma}{\sigma - 1} \right)^{-\sigma} \beta_{Fi} p_{Fi}^{-\sigma} \Rightarrow \pi_{Fi} = \frac{1}{\sigma - 1} \beta_{Fi} c_i^{1-\sigma} \quad (19)$$

Analogously given the foreign demand function for domestic intermediates, we derive export profit of the intermediate producer:

$$q_{FID} = \left( \frac{\rho}{\rho - 1} \right)^{-\rho} \beta_{FID} p_{FID}^{-\rho} \Rightarrow \pi_{FID} = \frac{1}{\rho - 1} \beta_{FID} w^{1-\rho} \quad (20)$$

Domestic profits depend on endogenous markups  $\mu_{Hi}$  that firm charge on the domestic market for the final good. To derive them, consider the expression for price index:

$$P = \left[ \sum_{i=1}^N \beta_i \mu_{Hi}^{1-\sigma} c_i^{1-\sigma} + \beta_F (\tau_F)^{1-\sigma} \right]^{\frac{1}{1-\sigma}} \quad (21)$$

Each markup is given implicitly by the following equation, which has a unique solution on  $[1, +\infty)$ :

$$\mu_{Hi} - \left( \frac{\beta_i c_i^{1-\sigma}}{P^{1-\sigma}} \right) \mu_{Hi}^{2-\sigma} = \frac{\sigma}{\sigma - 1} \quad (22)$$

Once we solve equation for each markup, price index is simply a function of domestic wage and model parameters.

Then we can write profits of the final good producers on the domestic market:

$$\pi_{Hi} = (\mu_{Hi} - 1) \underbrace{\left( \frac{\beta_i \mu_{Hi}^{-\sigma} c_i^{1-\sigma}}{P^{1-\sigma}} \right) I}_{\text{Domestic cost of } i} \quad (23)$$

For the intermediate firm:

$$\pi_{HID} = \sum_{i=1}^N \frac{1}{\rho} \underbrace{\left( \frac{1 - \alpha}{1 + \tau_I^{1-\rho} \alpha_{Fi}^\rho w^{\rho-1}} \right)}_{\text{Domestic cost share of } i} \underbrace{\left[ \left( \frac{\beta_i \mu_{Hi}^{-\sigma} c_i^{1-\sigma}}{P^{1-\sigma}} \right) I + \beta_{Fi} c_i^{1-\sigma} \right]}_{\text{Total cost of } i} \quad (24)$$

We can now come back to the equality:

$$I = \sum_{i=1}^N \pi_i + \pi_{ID} + w = \sum_{i=1}^N \pi_{Hi} + \sum_{i=1}^N \pi_{Fi} + \pi_{HID} + \pi_{FID} + w \quad (25)$$

This equality allows us to solve for the income as a function of wage and parameters of the model:

$$I = \frac{\left( \sum_{i=1}^N \left( \frac{1}{\sigma-1} + \frac{1}{\rho} \frac{1-\alpha}{1+\tau_I^{1-\rho} \alpha_{Fi}^\rho w^{\rho-1}} \right) \beta_{Fi} c_i^{1-\sigma} \right) + \frac{1}{\rho-1} \beta_{FID} w^{1-\rho} + w}{1 - \left( \sum_{i=1}^N \left( \mu_{Hi} - 1 + \frac{1}{\rho} \frac{1-\alpha}{1+\tau_I^{1-\rho} \alpha_{Fi}^\rho w^{\rho-1}} \right) \frac{\beta_i \mu_{Hi}^{-\sigma} c_i^{1-\sigma}}{p^{1-\sigma}} \right)} \quad (26)$$

Next, use trade balance equation:

$$\underbrace{\frac{\sigma}{\sigma-1} \sum_{i=1}^N \beta_{Fi} c_i^{1-\sigma} + \frac{\rho}{\rho-1} \beta_{FID} w^{1-\rho}}_{\text{Exports}} = \underbrace{\sum_{i=1}^N \left( \frac{(1-\alpha) \tau_I^{1-\rho} \alpha_{Fi}^\rho w^{\rho-1}}{1 + \tau_I^{1-\rho} \alpha_{Fi}^\rho w^{\rho-1}} \right) \left[ \left( \frac{\beta_i \mu_{Hi}^{-\sigma} c_i^{1-\sigma}}{p^{1-\sigma}} \right) I + \beta_{Fi} c_i^{1-\sigma} \right] + \left( \frac{\beta_F (\tau_F)^{1-\sigma}}{p^{1-\sigma}} \right) I}_{\text{Imports}} \quad (27)$$

## C.4 Algorithm for Simulation (Many Importers)

1. Set number of the domestic final good producers  $N$ .
2. Set parameter values (  $\underbrace{\sigma, \rho}_{\text{elasticities}} \quad \underbrace{\alpha, \beta_{FID}}_{\text{intermediate good}} \quad \underbrace{\tau_F, \tau_I, \beta_F}_{\text{foreign goods}}$  )
3. Generate (correlated) vector of variables  $(\beta_i, \beta_{Fi}, \alpha_{Fi}, \phi_i)$  for each of the domestic final good producers.
4. Assume value for  $w$ .
5. Derive marginal cost for each firm  $c_i$  using equation (1).
6. Assume value for the price index and then solve for each  $\mu_i$  using equation (5).

Iterate until equation (4) holds as equality.

7. Derive profits using equations (2)-(3) and (6)-(7). Combine them using equation (8) to derive income.

8. Derive error from trade balance using equation (9).

9. Iterate wage until error from trade balance is equal to 0.

## C.5 Simulation Results (Many Importers)

**Parameters of simulation:** There are 25 domestic final good producers. The parameters for the simulation are given by the following:  $\sigma = 5, \rho = 3, \alpha = 0.5, \beta_{FID} = 1000$ , and  $\beta_F = 0.1$ . Trade liberalization is given by changing trade costs from  $\tau_F = \tau_I = 1.25$  to  $\tau_F = \tau_I = 1$ . Firm-specific variables are generated in following way:  $\beta_i = \exp(x_{1i})$ ,  $\beta_{Fi} = \exp(x_{2i})$ ,  $\alpha_{Fi} = 0.05 \times \exp(x_{3i})$ ,  $\phi_i = \exp(x_{4i})$ , where

$$X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \sim N \left( \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 & c & c & c \\ c & 1 & c & c \\ c & c & 1 & c \\ c & c & c & 1 \end{pmatrix} \right)$$

**Results:** Let's present results for different levels of correlation parameter  $c$  and different types of trade liberalizations.

Table B.1 shows the changes in aggregate (cost-weighted) domestic markup:

$$M_d = \frac{\sum_{i=1}^{25} \beta_i \mu_{Hi}^{1-\sigma} c_i^{1-\sigma}}{\sum_{i=1}^{25} \beta_i \mu_{Hi}^{-\sigma} c_i^{1-\sigma}} \quad (28)$$

We could see that liberalizing final goods' tariffs generally has pro-competitive effects, while liberalizing intermediate goods' tariffs generally has anti-competitive effects. Both effects intensify as correlation parameter increases — for final tariff this is driven by fact that market concentration increases with higher  $c$  (firms with better productivity shocks also have strong advantage in access to foreign inputs and grow

larger), while for input tariffs it combines direct effect of larger firms benefiting more from input tariff cuts with the effect of increased concentration.<sup>31</sup>

| $\tau/c$                       | $c = 0$ | $c = 0.25$ | $c = 0.5$ | $c = 0.75$ |
|--------------------------------|---------|------------|-----------|------------|
| $\tau_F = 1.25, \tau_I = 1.25$ | 1.2998  | 1.5949     | 1.6866    | 1.8766     |
| $\tau_F = 1, \tau_I = 1.25$    | 1.2786  | 1.5162     | 1.5925    | 1.7763     |
| $\tau_F = 1.25, \tau_I = 1$    | 1.3039  | 1.6806     | 1.7561    | 1.9407     |
| $\tau_F = 1, \tau_I = 1$       | 1.2809  | 1.5897     | 1.6544    | 1.8327     |

Table B.1: Simulation Results - Aggregate Markup

The following Table B.2 presents results for real wages.

| $\tau/c$                       | $c = 0$ | $c = 0.25$ | $c = 0.5$ | $c = 0.75$ |
|--------------------------------|---------|------------|-----------|------------|
| $\tau_F = 1.25, \tau_I = 1.25$ | 5.3085  | 5.7149     | 6.8177    | 7.8127     |
| $\tau_F = 1, \tau_I = 1.25$    | 6.1676  | 6.6008     | 7.6337    | 8.7215     |
| $\tau_F = 1.25, \tau_I = 1$    | 5.3436  | 5.8501     | 7.1701    | 8.4218     |
| $\tau_F = 1, \tau_I = 1$       | 6.1944  | 6.7563     | 7.9930    | 9.3619     |

Table B.2: Simulation Results - Real Wage

Finally, Table B.3 presents results for the total welfare (real income).

<sup>31</sup>Interestingly, while aggregate markup decreases in case  $c=0.5$  under liberalization, sum of competitive effects of tariffs on the real wage is slightly negative --- this is likely driven by increased variance of markups that decreases real wages (similar to Epifani and Gancia 2011, Edmond et al. 2019).

| $\tau/c$                       | $c = 0$ | $c = 0.25$ | $c = 0.5$ | $c = 0.75$ |
|--------------------------------|---------|------------|-----------|------------|
| $\tau_F = 1.25, \tau_I = 1.25$ | 8.0370  | 9.3733     | 12.0160   | 14.7503    |
| $\tau_F = 1, \tau_I = 1.25$    | 9.2757  | 10.3966    | 12.6324   | 15.2422    |
| $\tau_F = 1.25, \tau_I = 1$    | 8.1006  | 9.8534     | 13.0669   | 16.3729    |
| $\tau_F = 1, \tau_I = 1$       | 9.3200  | 10.8442    | 13.5684   | 16.7340    |

Table B.3: Simulation Results - Real Income

From the tables we can see that as  $c$  increases gains from intermediate input tariff cuts, as increasing  $c$  leads to higher total share of imported inputs — this is a separate effect from considering effects on competition. One could also see that higher  $c$  and  $\tau_F$ , and lower  $\tau_I$  leads to lower share of real wage in real income.

**Decomposition exercise:** To explain the connection between competitive effects (markup table) and effects on the real wages/incomes, we decompose changes in real wage from full liberalization (moving from  $\tau_F = \tau_I = 1.25$  to  $\tau_F = \tau_I = 1$ ) into three first-order terms and a remainder term for cases  $c = 0$  and  $c = 0.5$ . Table B.4 shows the decomposition results.

| Effect on Real Wage     | $c = 0$ | $c = 0.25$ | $c = 0.5$ | $c = 0.75$ |
|-------------------------|---------|------------|-----------|------------|
| Direct Cost Effect      | 0.1418  | 0.1484     | 0.1320    | 0.1443     |
| Pro-Competitive Effect  | 0.0050  | 0.0284     | 0.0410    | 0.0567     |
| Anti-Competitive Effect | -0.0004 | -0.0154    | -0.0212   | -0.0268    |
| Second-Order effects    | 0.0079  | 0.0059     | -0.0018   | 0.0067     |
| Total Effect            | 0.1543  | 0.1673     | 0.1590    | 0.1809     |

Table B.4: Simulation Results - Welfare Decomposition

We see that knowing  $c$  parameter affects our interpretation of pro/anti-competitive effects of trade. Competitive effects are generally more pronounced once  $c$  is higher as market is more concentrated. Note, however, that competitive effects of input tariffs almost completely disappear under  $c = 0$ , while becoming comparable in size (albeit smaller) to competitive effects of output tariffs under  $c > 0$ . We can also see that first-order decomposition provides a reasonably good approximation to the total effect on real wages (e.g. second-order effects are small).

## D Robustness

### D.1 Alternative Market Structures

We proceed by redoing the above welfare analysis for alternative market structures. While the baseline model is derived assuming quantity competition (Cournot), we now consider the alternative price competition (Bertrand). Under the Bertrand price competition with Cobb-Douglas industry aggregator,  $\Gamma_{i,t} = \Gamma_{-i,t} = \frac{(\sigma-1)s_{i,t}}{1+(\sigma-1)(1-s_{i,t})}$ , which is still a monotonically increasing function of  $s_i$ .

$$dp_i = \frac{1 - s_i}{1 - s_i + \frac{(\sigma-1)s_i}{1+(\sigma-1)(1-s_i)}} dmc_i + \frac{\frac{(\sigma-1)s_i}{1+(\sigma-1)(1-s_i)}}{1 - s_i + \frac{(\sigma-1)s_i}{1+(\sigma-1)(1-s_i)}} d \ln P \quad (29)$$

We proceed with the quantitative analysis following the same method as described in section 2.3.

### D.2 Aggregation across Sectors

We extend our welfare decomposition to a multi-sector environment. When assuming Cobb-Douglas industry aggregator we have:

$$\begin{aligned}
d\ln(w) - d\ln(P) &= \sum_s \phi_s [d\ln(w_s) - d\ln(P_s)] \\
&= \underbrace{\sum_s \phi_s \left[ s_{f,s} (d\ln w_s - d\tau_s) + \sum_{i=1}^{N(s)} s_i s_i^I (d\ln(w_s) - d\tau_i^{input}) \right]}_{\text{Gains without competitive effects}} \\
&\quad + \underbrace{\sum_s \phi_s \left[ s_{f,s} (d\ln w_s - d\tau_s) \frac{\sum_{i=1}^{N(s)} \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}}{1 - \sum_{i=1}^{N(s)} \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}} \right]}_{\text{Pro-competitive effects}} \\
&\quad - \underbrace{\sum_s \phi_s \left[ \frac{Cov \left( s_i^I, \frac{(\sigma-1)s_i}{1+(\sigma-2)s_i}; s_i (d\ln w_s - d\tau_i^{input}) \right)}{1 - \sum_{i=1}^{N(s)} \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}} \right]}_{\text{Anti-competitive effects}} \tag{30}
\end{aligned}$$

In the model with multiple sectors, the overall anti-competitive effects is now determined by  $-\sum_s \phi_s \left[ \frac{Cov \left( s_i^I, \frac{(\sigma-1)s_i}{1+(\sigma-2)s_i}; s_i (d\ln w - d\tau_i^{input}) \right)}{1 - \sum_{i=1}^{N(s)} \frac{(\sigma-1)s_i^2}{1+(\sigma-2)s_i}} \right]$ , which is a weighted average of the weighted covariance between firm market share and import share across all industries, where the weights are the Cobb-Douglas coefficients.